Data Mining:

Concepts and Techniques

(3rd ed.)

— Chapter 10 —

Jiawei Han, Micheline Kamber, and Jian Pei
University of Illinois at Urbana-Champaign &
Simon Fraser University
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Chapter 10. Cluster Analysis: Basic Concepts and Methods

- Cluster Analysis: Basic Concepts
- Partitioning Methods
- Hierarchical Methods
- Density-Based Methods
- Grid-Based Methods
- Evaluation of Clustering
- Summary

What is Cluster Analysis?

- Cluster: A collection of data objects
  - similar (or related) to one another within the same group
  - dissimilar (or unrelated) to the objects in other groups
- Cluster analysis (or *clustering*, *data segmentation, …*)
  - Finding similarities between data according to the characteristics found in the data and grouping similar data objects into clusters
- Unsupervised learning: no predefined classes (i.e., *learning by observations* vs. learning by examples: supervised)
- Typical applications
  - As a stand-alone tool to get insight into data distribution
  - As a preprocessing step for other algorithms

Clustering for Data Understanding and Applications

- Biology: taxonomy of living things: kingdom, phylum, class, order, family, genus and species
- Information retrieval: document clustering
- Land use: Identification of areas of similar land use in an earth observation database
- Marketing: Help marketers discover distinct groups in their customer bases, and then use this knowledge to develop targeted marketing programs
- City-planning: Identifying groups of houses according to their house type, value, and geographical location
- Earth-quake studies: Observed earth quake epicenters should be clustered along continent faults
- Climate: understanding earth climate, find patterns of atmospheric and ocean
- Economic Science: market resarch

Clustering as a Preprocessing Tool (Utility)

- Summarization:
  - Preprocessing for regression, PCA, classification, and association analysis
- Compression:
  - Image processing: vector quantization
- Finding K-nearest Neighbors
  - Localizing search to one or a small number of clusters
- Outlier detection
  - Outliers are often viewed as those “far away” from any cluster

Considerations for Cluster Analysis

- Partitioning criteria
  - Single level vs. hierarchical partitioning (often, multi-level hierarchical partitioning is desirable)
- Separation of clusters
  - Exclusive (e.g., one customer belongs to only one region) vs. non-exclusive (e.g., one document may belong to more than one class)
- Similarity measure
  - Distance-based (e.g., Euclidian, road network, vector) vs. connectivity-based (e.g., density or contiguity)
- Clustering space
  - Full space (often when low dimensional) vs. subspaces (often in high-dimensional clustering)

Requirements and Challenges

- Scalability
  - Clustering all the data instead of only on samples
- Ability to deal with different types of attributes
  - Numerical, binary, categorical, ordinal, linked, and mixture of these
- Constraint-based clustering
  - User may give inputs on constraints
  - Use domain knowledge to determine input parameters
- Interpretability and usability
- Others
  - Discovery of clusters with arbitrary shape
  - Ability to deal with noisy data
  - Incremental clustering and insensitivity to input order
  - High dimensionality

Major Clustering Approaches (I)

- Partitioning approach:
  - Construct various partitions and then evaluate them by some criterion, e.g., minimizing the sum of square errors
  - Typical methods: k-means, k-medoids, CLARANS
- Hierarchical approach:
  - Create a hierarchical decomposition of the set of data (or objects) using some criterion
  - Typical methods: Diana, Agnes, BIRCH, CAMELEON
- Density-based approach:
  - Based on connectivity and density functions
  - Typical methods: DBSACN, OPTICS, DenClue
- Grid-based approach:
  - based on a multiple-level granularity structure
  - Typical methods: STING, WaveCluster, CLIQUE

Major Clustering Approaches (II)

- Model-based:
  - A model is hypothesized for each of the clusters and tries to find the best fit of that model to each other
  - Typical methods: EM, SOM, COBWEB
- Frequent pattern-based:
  - Based on the analysis of frequent patterns
  - Typical methods: p-Cluster
- User-guided or constraint-based:
  - Clustering by considering user-specified or application-specific constraints
  - Typical methods: COD (obstacles), constrained clustering
- Link-based clustering:
  - Objects are often linked together in various ways
  - Massive links can be used to cluster objects: SimRank, LinkClus

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Partitioning Algorithms: Basic Concept

- <u>Partitioning method:</u> Partitioning a database ***D*** of ***n*** objects into a set of ***k*** clusters, such that the sum of squared distances is minimized (where c_i is the centroid or medoid of cluster C_i)

$$E = \Sigma_{i=1}^{k} \Sigma_{p \in C_i} (p - c_i)^2$$

- Given *k*, find a partition of *k clusters* that optimizes the chosen partitioning criterion
  - Global optimal: exhaustively enumerate all partitions
  - Heuristic methods: *k-means* and *k-medoids* algorithms
  - <u>*k-means*</u> (MacQueen'67, Lloyd'57/'82): Each cluster is represented by the center of the cluster
  - <u>*k-medoids*</u> or PAM (Partition around medoids) (Kaufman & Rousseeuw'87): Each cluster is represented by one of the objects in the cluster

The *K-Means* Clustering Method

- Given *k*, the *k-means* algorithm is implemented in four steps:
  - Partition objects into *k* nonempty subsets
  - Compute seed points as the centroids of the clusters of the current partitioning (the centroid is the center, i.e., *mean point*, of the cluster)
  - Assign each object to the cluster with the nearest seed point
  - Go back to Step 2, stop when the assignment does not change

K-Means Clustering – Solved Example

- Suppose that the data mining task is to cluster points into three clusters,
- where the points are
- $A1(2, 10)$, $A2(2, 5)$, $A3(8, 4)$, $B1(5, 8)$, $B2(7, 5)$, $B3(6, 4)$, $C1(1, 2)$, $C2(4, 9)$.
- The distance function is Euclidean distance.
- Suppose initially we assign $A1$, $B1$, and $C1$ as the center of each cluster, respectively.

K-Means Clustering – Solved Example

Initial Centroids:
A1: (2, 10)
B1: (5, 8)
C1: (1, 2)

Data Points			Distance to						Cluster	New Cluster
			2	10	5	8	1	2		
A1	2	10	0.00		3.61		8.06		1	
A2	2	5	5.00		4.24		3.16		3	
A3	8	4	8.49		5.00		7.28		2	
B1	5	8	3.61		0.00		7.21		2	
B2	7	5	7.07		3.61		6.71		2	
B3	6	4	7.21		4.12		5.39		2	
C1	1	2	8.06		7.21		0.00		3	
C2	4	9	2.24		1.41		7.62		2	

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Initial Centroids:
A1: (2, 10)
B1: (5, 8)
C1: (1, 2)

New Centroids:
A1: (2, 10)
B1: (6, 6)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			2	10	5	8	1	2		
A1	2	10	0.00		3.61		8.06		1	
A2	2	5	5.00		4.24		3.16		3	
A3	8	4	8.49		5.00		7.28		2	
B1	5	8	3.61		0.00		7.21		2	
B2	7	5	7.07		3.61		6.71		2	
B3	6	4	7.21		4.12		5.39		2	
C1	1	2	8.06		7.21		0.00		3	
C2	4	9	2.24		1.41		7.62		2	

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Current Centroids:
A1: (2, 10)
B1: (6, 6)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			2	10	6	6	1.5	1.5		
A1	2	10	0.00		5.66		6.52		1	1
A2	2	5	5.00		4.12		1.58		3	3
A3	8	4	8.49		2.83		6.52		2	2
B1	5	8	3.61		2.24		5.70		2	2
B2	7	5	7.07		1.41		5.70		2	2
B3	6	4	7.21		2.00		4.53		2	2
C1	1	2	8.06		6.40		1.58		3	3
C2	4	9	2.24		3.61		6.04		2	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Current Centroids:
A1: (2, 10)
B1: (6, 6)
C1: (1.5, 3.5)

New Centroids:
A1: (3, 9.5)
B1: (6.5, 5.25)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			2	10	6	6	1.5	1.5		
A1	2	10	0.00		5.66		6.52		1	1
A2	2	5	5.00		4.12		1.58		3	3
A3	8	4	8.49		2.83		6.52		2	2
B1	5	8	3.61		2.24		5.70		2	2
B2	7	5	7.07		1.41		5.70		2	2
B3	6	4	7.21		2.00		4.53		2	2
C1	1	2	8.06		6.40		1.58		3	3
C2	4	9	2.24		3.61		6.04		2	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Current Centroids:
A1: (3, 9.5)
B1: (6.5, 5.25)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			3	9.5	6.5	5.25	1.5	3.5		
A1	2	10	1.12		6.54		6.52		1	1
A2	2	5	4.61		4.51		1.58		3	3
A3	8	4	7.43		1.95		6.52		2	2
B1	5	8	2.50		3.13		5.70		2	1
B2	7	5	6.02		0.56		5.70		2	2
B3	6	4	6.26		1.35		4.53		2	2
C1	1	2	7.76		6.39		1.58		3	3
C2	4	9	1.12		4.51		6.04		1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-Means Clustering – Solved Example

Current Centroids:
A1: (3, 9.5)
B1: (6.5, 5.25)
C1: (1.5, 3.5)

New Centroids:
A1: (3.67, 9)
B1: (7, 4.33)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			3	9.5	6.5	5.25	1.5	3.5		
A1	2	10	1.12		6.54		6.52		1	1
A2	2	5	4.61		4.51		1.58		3	3
A3	8	4	7.43		1.95		6.52		2	2
B1	5	8	2.50		3.13		5.70		2	1
B2	7	5	6.02		0.56		5.70		2	2
B3	6	4	6.26		1.35		4.53		2	2
C1	1	2	7.76		6.39		1.58		3	3
C2	4	9	1.12		4.51		6.04		1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

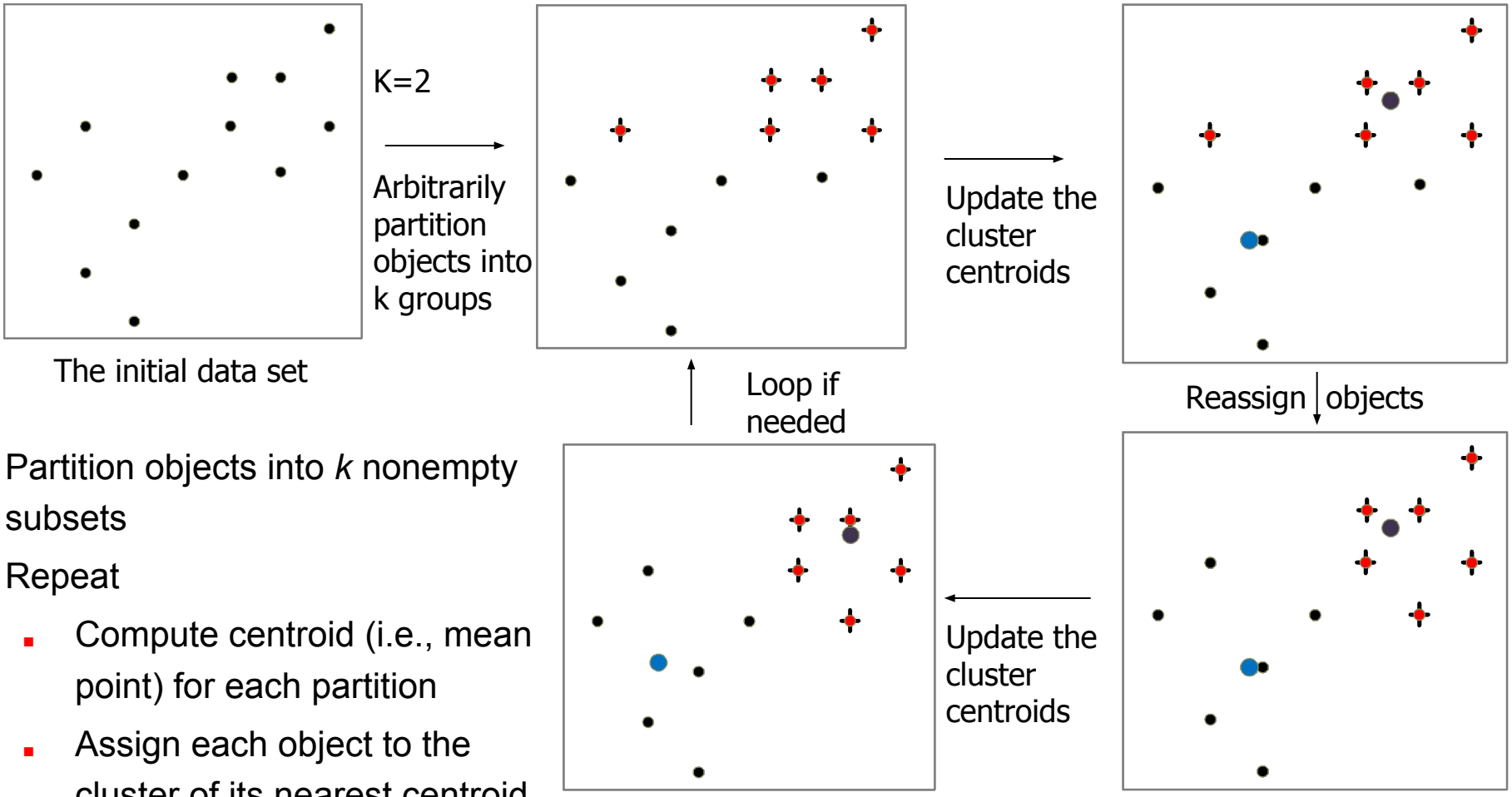
K-Means Clustering – Solved Example

Current Centroids:
A1: (3.67, 9)
B1: (7, 4.33)
C1: (1.5, 3.5)

Data Points			Distance to						Cluster	New Cluster
			3.67	9	7	4.33	1.5	3.5		
A1	2	10	1.94		7.56		6.52		1	1
A2	2	5	4.33		5.04		1.58		3	3
A3	8	4	6.62		1.05		6.52		2	2
B1	5	8	1.67		4.18		5.70		1	1
B2	7	5	5.21		0.67		5.70		2	2
B3	6	4	5.52		1.05		4.53		2	2
C1	1	2	7.49		6.44		1.58		3	3
C2	4	9	0.33		5.55		6.04		1	1

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

An Example of *K-Means* Clustering



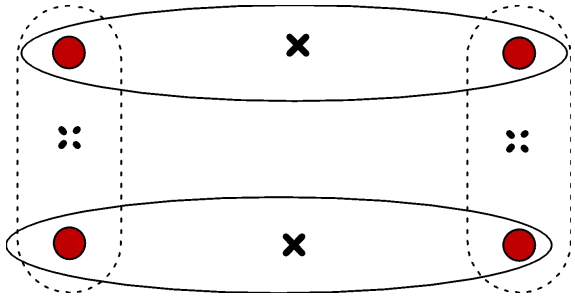
- Partition objects into *k* nonempty subsets
- Repeat
  - Compute centroid (i.e., mean point) for each partition
  - Assign each object to the cluster of its nearest centroid
- Until no change

Comments on the *K-Means* Method

- <u>Strength:</u> *Efficient*: *O*(*tkn*), where *n* is # objects, *k* is # clusters, and *t* is # iterations. Normally, *k*, *t* << *n*.
  - Comparing: PAM: $O(k(n-k)^2)$, CLARA: $O(ks^2 + k(n-k))$
- <u>Comment:</u> Often terminates at a *local optimal*.
- <u>Weakness</u>
  - Applicable only to objects in a continuous n-dimensional space
    - Using the k-modes method for categorical data
    - In comparison, k-medoids can be applied to a wide range of data
  - Need to specify *k,* the *number* of clusters, in advance (there are ways to automatically determine the best k (see Hastie et al., 2009)
  - Sensitive to noisy data and *outliers*
  - Not suitable to discover clusters with *non-convex shapes*

Variations of the *K-Means* Method

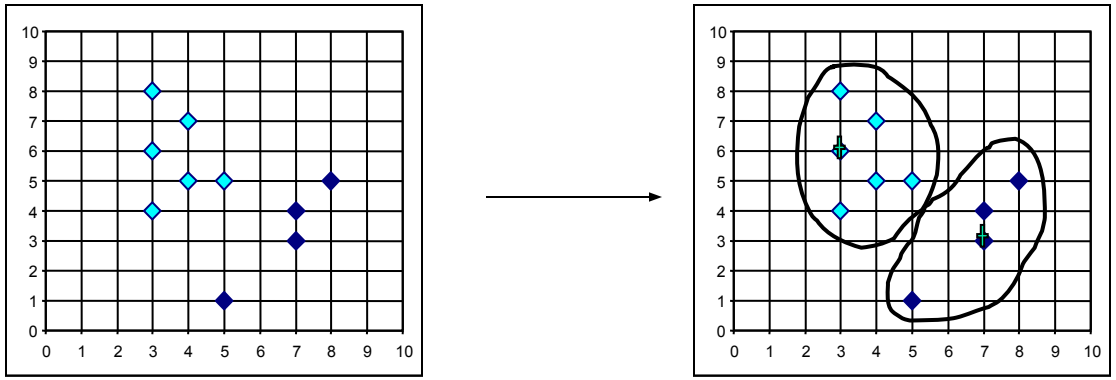
- Most of the variants of the *k-means* which differ in
  - Selection of the initial *k* means
  - Dissimilarity calculations
  - Strategies to calculate cluster means



- Handling categorical data: *k-modes*
  - Replacing means of clusters with <u>modes</u>
  - Using new dissimilarity measures to deal with categorical objects
  - Using a <u>frequency</u>-based method to update modes of clusters
  - A mixture of categorical and numerical data: *k-prototype* method

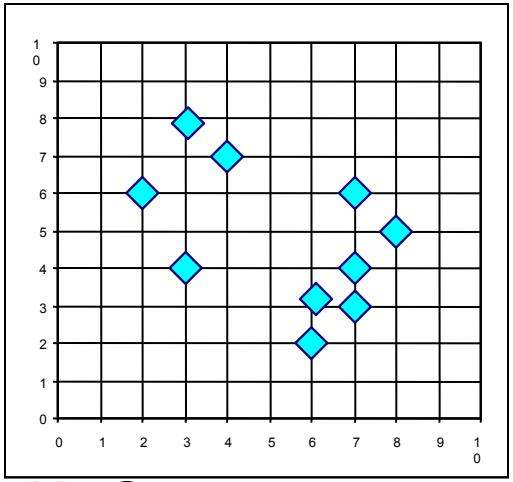
What Is the Problem of the K-Means Method?

- The k-means algorithm is sensitive to outliers !
  - Since an object with an extremely large value may substantially distort the distribution of the data
- K-Medoids: Instead of taking the **mean** value of the object in a cluster as a reference point, **medoids** can be used, which is the **most centrally located** object in a cluster



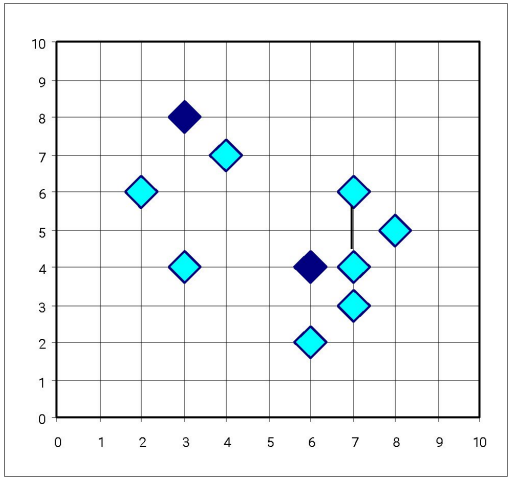
PAM: A Typical K-Medoids Algorithm

Total Cost = 20

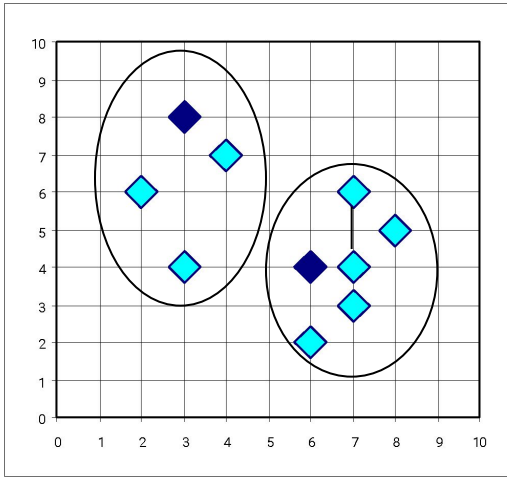


K=2

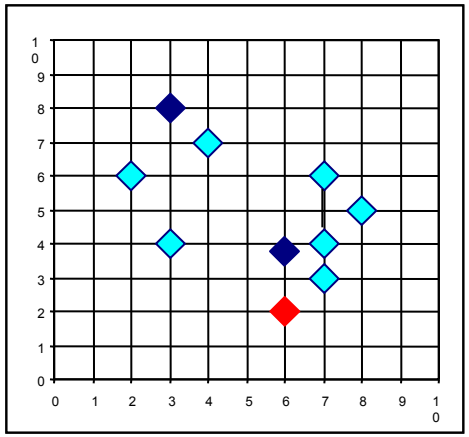
Arbitrary choose k object as initial medoids



Assign each remaining object to nearest medoids

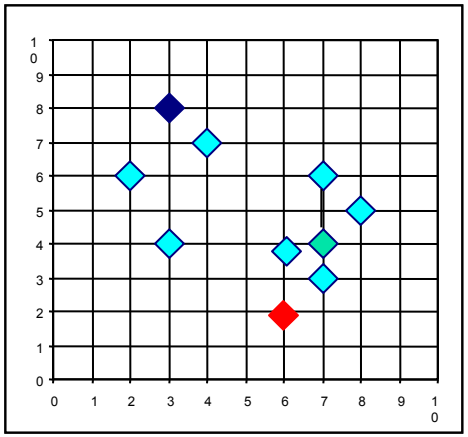


Randomly select a nonmedoid object, O_{ramdom}



Compute total cost of swapping

Total Cost = 26



Swapping O and O_{ramdom}

If quality is improved.

Do loop

Until no change

The K-Medoid Clustering Method

- *K-Medoids* Clustering: Find *representative* objects (medoids) in clusters
  - *PAM* (Partitioning Around Medoids, Kaufmann & Rousseeuw 1987)
    - Starts from an initial set of medoids and iteratively replaces one of the medoids by one of the non-medoids if it improves the total distance of the resulting clustering
    - *PAM* works effectively for small data sets, but does not scale well for large data sets (due to the computational complexity)
- Efficiency improvement on PAM
  - *CLARA* (Kaufmann & Rousseeuw, 1990): PAM on samples
  - *CLARANS* (Ng & Han, 1994): Randomized re-sampling

i	x	y
X1	2	6
X2	3	4
X3	3	8
X4	4	7
X5	6	2
X6	6	4
X7	7	3
X8	7	4
X9	8	5
X10	7	6

- Apply K-Medoid clustering algorithm to form two clusters.
- Use Manhattan distance to find the between data point and medoid.

Step 1

- Select two medoids
- C1=(3, 4)
- C2=(7, 4)
- $Manhattan\ Dist = |x_1 - x_2| + |y_1 - y_2|$
- $Mdist[(2,6),(3,4)] = |2-3| + |6-4| = 3$
- $Mdist[(3,4),(3,4)] = |3-3| + |4-4| = 0$

i	x	y	C1	C2	Cluster
X1	2	6	3		
X2	3	4	0		
X3	3	8			
X4	4	7			
X5	6	2			
X6	6	4			
X7	7	3			
X8	7	4			
X9	8	5			
X10	7	6			

Step 1

- Select two medoids
- C1=(3, 4)
- C2=(7, 4)
- $Manhattan\ Dist = |x_1 - x_2| + |y_1 - y_2|$
- $Mdist[(2,6),(3,4)] = |2-3| + |6-4| = 3$
- $Mdist[(3,4),(3,4)] = |3-3| + |4-4| = 0$

i	x	y	C1	C2	Cluster
X1	2	6	3	7	
X2	3	4	0	4	
X3	3	8	4	8	
X4	4	7	4	6	
X5	6	2	5	3	
X6	6	4	3	1	
X7	7	3	5	1	
X8	7	4	4	0	
X9	8	5	6	2	
X10	7	6	6	2	

Step 1

- Select two medoids
- C1=(3, 4)
- C2=(7, 4)
- $Manhattan\ Dist = |x_1 - x_2| + |y_1 - y_2|$
- $Mdist[(2,6),(3,4)] = |2-3| + |6-4| = 3$
- $Mdist[(3,4),(3,4)] = |3-3| + |4-4| = 0$

i	x	y	C1	C2	Cluster
X1	2	6	3	7	C1
X2	3	4	0	4	C1
X3	3	8	4	8	C1
X4	4	7	4	6	C1
X5	6	2	5	3	C2
X6	6	4	3	1	C2
X7	7	3	5	1	C2
X8	7	4	4	0	C2
X9	8	5	6	2	C2
X10	7	6	6	2	C2

- C1: {(2,6), **(3,4),** (3,8), (4,7)}
- C2: {(6, 2), (6, 4), (7, 3), **(7, 4),** (8, 5), (7,6)}
- **Calculate the Total Cost**
- $\boldsymbol{Cost(c,x)} = \sum_i |\boldsymbol{c_i} - \boldsymbol{x_i}|$
- $Total\ Cost = \{Cost\big((3,4),(2,6)\big) + Cost\big((3,4),(3,8)\big) + Cost\big((3,4),(4,7)\big) + Cost\big((7,4),(6,2)\big) + Cost\big((7,4),(6,4)\big) + Cost\big((7,4),(7,3)\big) + Cost\big((7,4),(8,5)\big) + Cost\big((7,4),(7,6)\big)\}$
  - $\boldsymbol{Total\ Cost = 3 + 4 + 4 + 2 + 3 + 1 + 1 + 2 = 20}$

Step 3

- Randomly select one non-medoid point and recalculate the cost.
- C1=(3, 4) and C2=(7, 4)
- O=(7, 3)
- Swap C2 with O
- New Medoids
- C1=(3, 4) and O=(7, 3)

i	x	y	C1	O	Cluster
X1	2	6			
X2	3	4			
X3	3	8			
X4	4	7			
X5	6	2			
X6	6	4			
X7	7	3			
X8	7	4			
X9	8	5			
X10	7	6			

Step 3

- New Medoids
- C1=(3, 4) and O=(7, 3)
- $Manhattan\ Dist = |x_1 - x_2| + |y_1 - y_2|$
- $Mdist[(2,6),(7,3)] = |2-7| + |6-3| = 8$

i	x	y	C1	O	Cluster
X1	2	6	3	8	C1
X2	3	4	0	5	C1
X3	3	8	4	9	C1
X4	4	7	4	7	C1
X5	6	2	5	2	O
X6	6	4	3	2	O
X7	7	3	5	0	O
X8	7	4	4	1	O
X9	8	5	6	3	O
X10	7	6	6	3	O

- C1: {(2,6), **(3,4),** (3,8), (4,7)}
- O: {(6, 2), (6, 4), **(7, 3),** (7, 4), (8, 5), (7,6)}
- **Calculate the Total Cost**
- $\boldsymbol{Cost(c,x)} = \sum_i |\boldsymbol{c_i} - \boldsymbol{x_i}|$
- $Current\ Total\ Cost = \{Cost\big((3,4),(2,6)\big) + Cost\big((3,4),(3,8)\big) + Cost\big((3,4),(4,7)\big) + Cost\big((7,3),(6,2)\big) + Cost\big((7,3),(6,4)\big) + Cost\big((7,3),(7,4)\big) + Cost\big((7,3),(8,5)\big) + Cost\big((7,3),(7,6)\big)\}$
- $\boldsymbol{Current\ Total\ Cost = 3 + 4 + 4 + 2 + 2 + 1 + 3 + 3 = 22}$

Step 4

- Cost of Swapping of medoid C2 with O
- $S = Current\ Total\ Cost - Previous\ Total\ Cost$
- $S = 22 - 20 = 2 > 0$
- Hence Swapping C2 with O is not a good Idea.
- Final Medoids are C1 =(3, 4) and C2=(7, 4)
- Clusters are
- C1: {(2,6), (3,4), (3,8), (4,7)}
- C2: {(6, 2), (6, 4), (7, 3), (7, 4), (8, 5), (7,6)}

Chapter 10. Cluster Analysis: Basic Concepts and Methods

- Cluster Analysis: Basic Concepts
- Partitioning Methods
- Hierarchical Methods
- Density-Based Methods
- Grid-Based Methods
- Evaluation of Clustering
- Summary

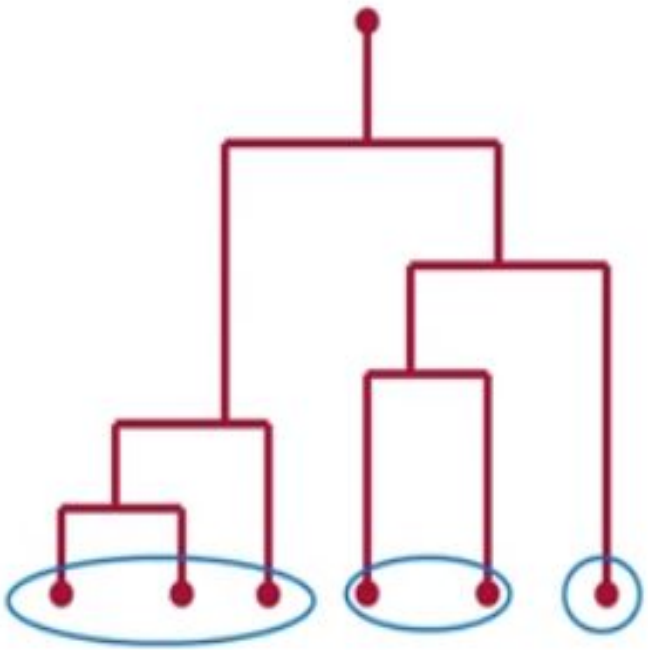
Hierarchical clustering

- Hierarchical clustering is an alternative approach to partitioning clustering for grouping objects based on their **similarity**.

- The groups are nested and organized as a tree

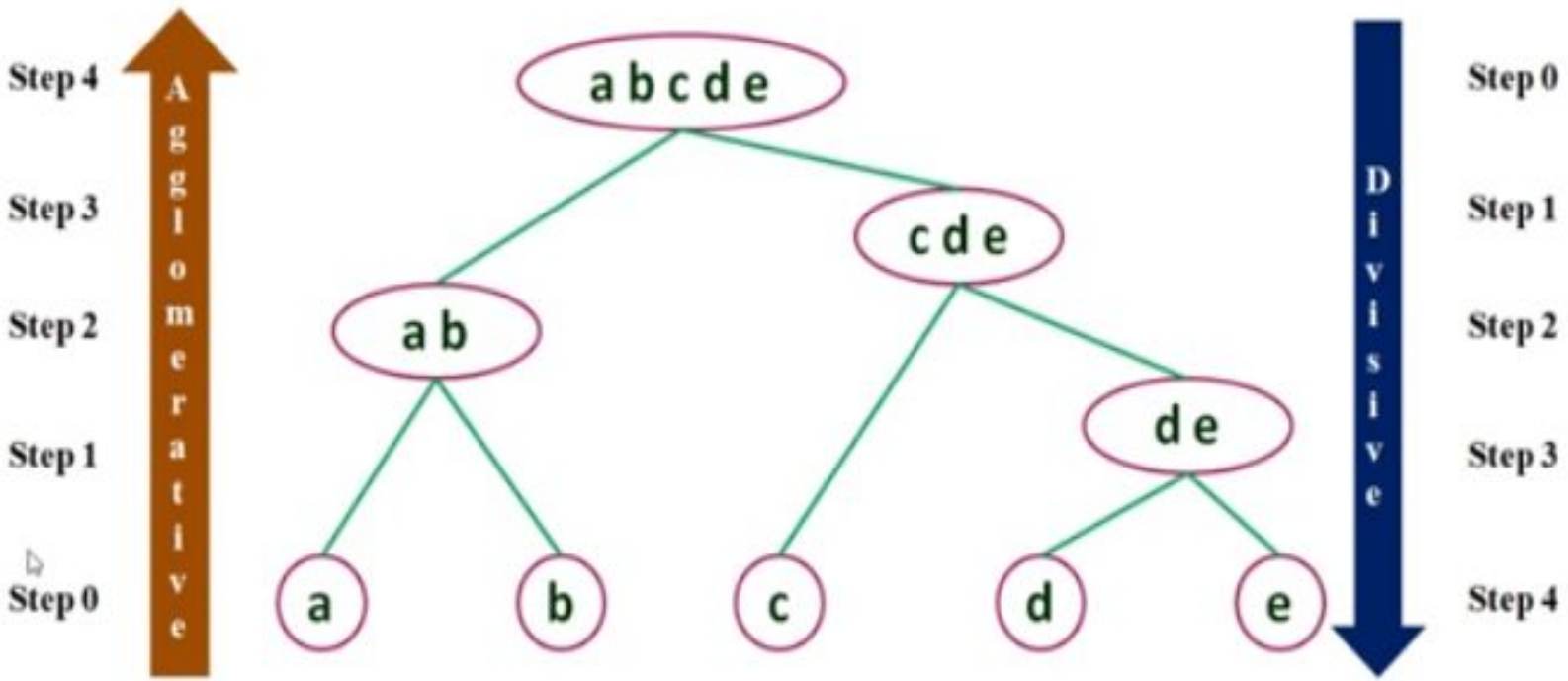
- Hierarchical clustering produces a sequence of clustering assignments.

- At one end, all points are in their own cluster, at the other end, all points are in one cluster



Two types of hierarchical clustering algorithms:

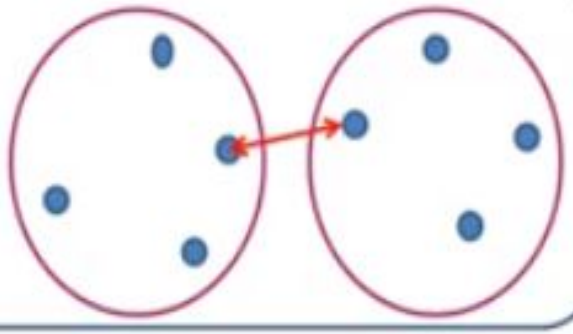
1. **Agglomerative** - Bottom up approach
2. **Divisive** - Top down approach



Cluster Distance Measures

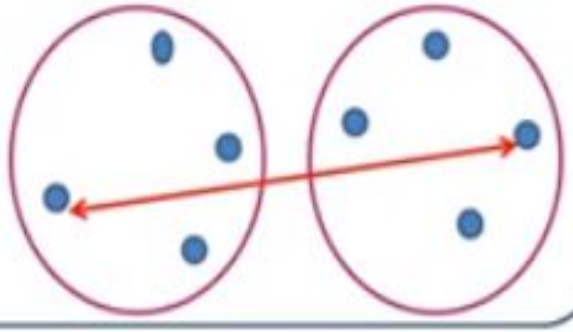
Single Linkage :

The distance between two clusters is the **shortest** distance between two points in each cluster.



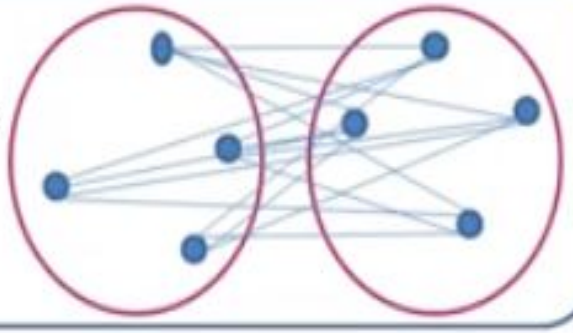
Complete Linkage :

The distance between two clusters is the **longest** distance between two points in each cluster.



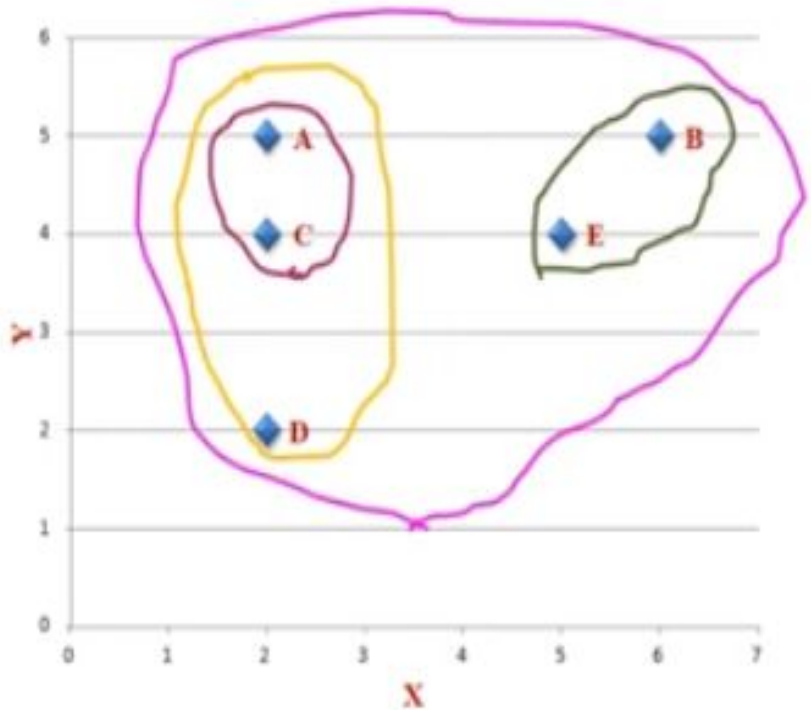
Average Linkage :

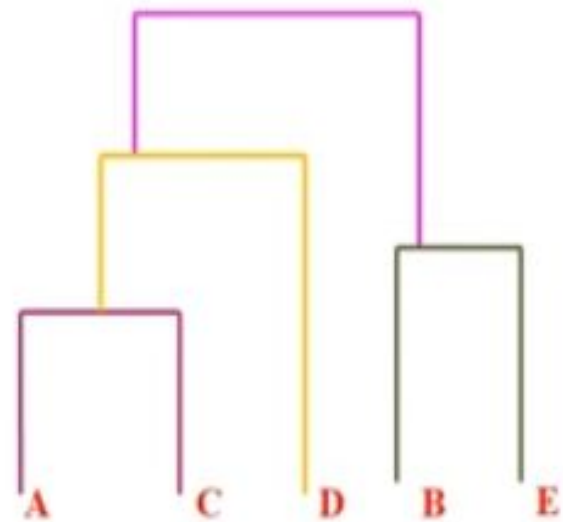
The distance between clusters is the **average** distance between each point in one cluster to every point in other cluster



Dendrogram:

A **Dendrogram** is a tree-like diagram that records the sequences of merges or split





Agglomerative - Bottom up approach

- Initially considered as a cluster of its own (leaf). Then, the most similar clusters are successively merged until there is only one cluster (root).

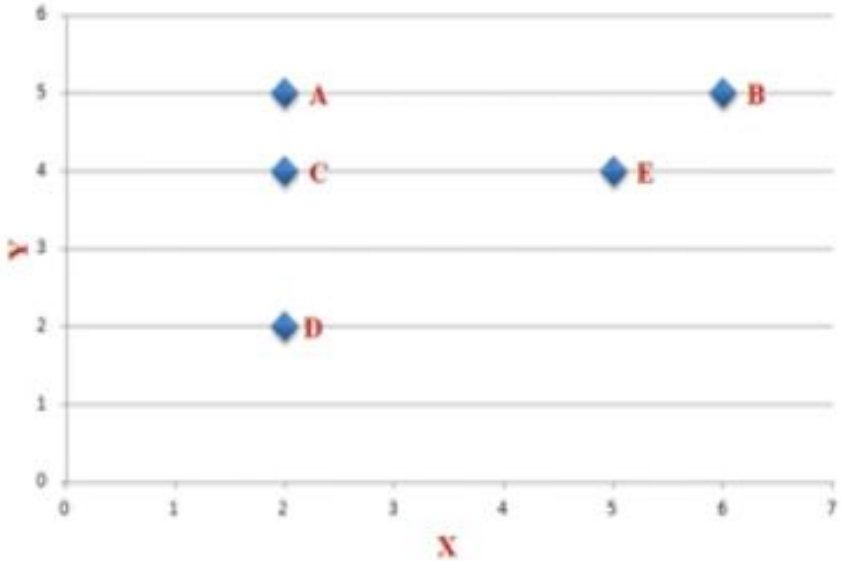
1). Convert all object features into a distance matrix.

2). Assign each object to a cluster so that for N objects we have N clusters.

3). Repeat until number of cluster into a single cluster. (Merge two closest clusters, Update "distance matrix")

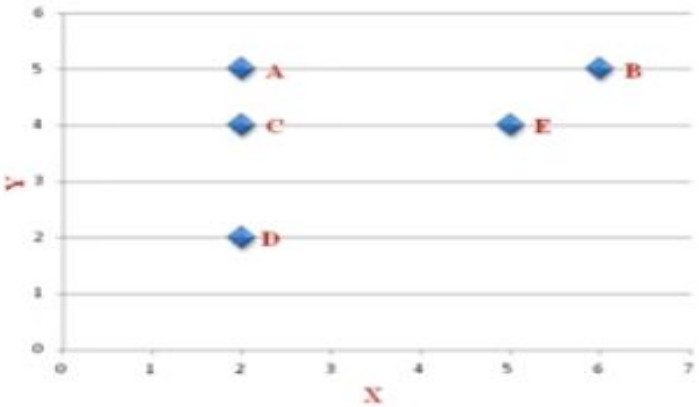
Example - AGNES
(AGglomerative NESting)

	X	Y
A	2	5
B	6	5
C	2	4
D	2	2
E	5	4



Distance Matrix

	A	B	C	D	E
A	0	4	1		
B					
C					
D					
E					



Euclidean distance : $d_{(x,y)} = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

A	2	5
B	6	5
C	2	4

$d_{(AB)} =$

$d = \sqrt{(6-2)^2 + (5-5)^2}$

$d = \sqrt{(4)^2 + (0)^2} \quad d = \sqrt{16+0}$

$d = \sqrt{16} \quad d = 4$

$d_{(AC)} =$

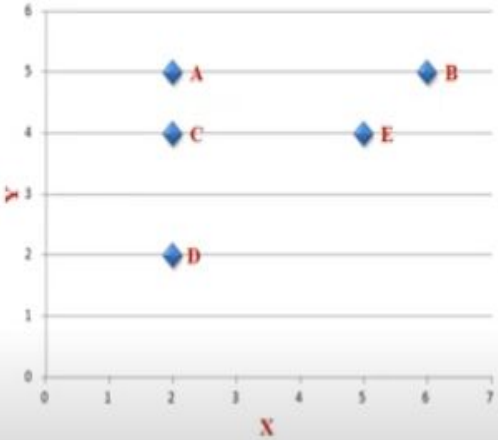
$d = \sqrt{(2-2)^2 + (4-5)^2}$

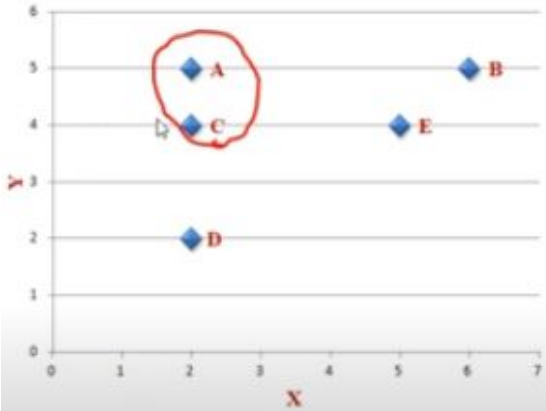
$d = \sqrt{(0)^2 + (-1)^2} \quad d = \sqrt{0+1}$

$d = \sqrt{1} \quad d = 1$

Distance Matrix

	A	B	C	D	E
A	0	4	1	3	3.16
B	4	0	4.12	5	1.41
C	1	4.12	0	2	3
D	3	5	2	0	3.6
E	3.16	1.41	3	3.6	0





Update distance matrix

	A,C	B	D	E
A,C	0	?	?	?
B	?	0	5	1.41
D	?	5	0	3.6
E	?	1.41	3.6	0

	A	B	C	D	E
A	0	4	1	3	3.16
B	4	0	4.12	5	1.41
C	1	4.12	0	2	3
D	3	5	2	0	3.6
E	3.16	1.41	3	3.6	0

$d_{(A,C)\rightarrow B} = \min(d_{AB}, d_{CB}) = \min(4, 4.12) = 4$

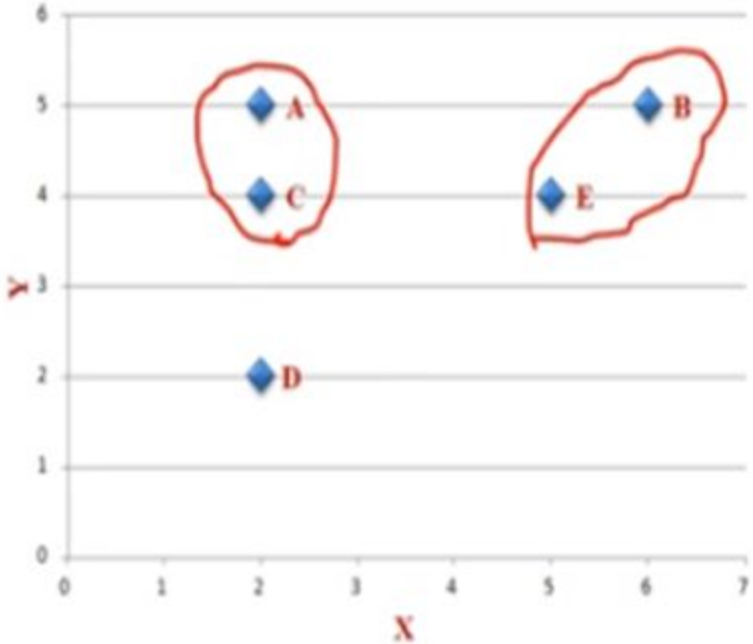
$d_{(A,C)\rightarrow D} = \min(d_{AD}, d_{CD}) = \min(3, 2) = 2$

$d_{(A,C)\rightarrow E} = \min(d_{AE}, d_{CE}) = \min(3.16, 3) = 3$

	A,C	B	D	E
A,C	0	4	2	3
B	4	0	5	1.41
D	2	5	0	3.6
E	3	1.41	3.6	0

Distance Matrix

	A,C	B	D	E
A,C	0	4	2	3
B	4	0	5	1.41
D	2	5	0	3.6
E	3	1.41	3.6	0



Update distance matrix

	A,C	B,E	D
A,C	0	?	2
B,E	?	0	?
D	2	?	0

	A	B	C	D	E
A	0	4	1	3	3.16
B	4	0	4.12	5	1.41
C	1	4.12	0	2	3
D	3	5	2	0	3.6
E	3.16	1.41	3	3.6	0

$d_{(A,C)\to(B,E)} = \min(d_{AB}, d_{AE}, d_{CB}, d_{CE})$

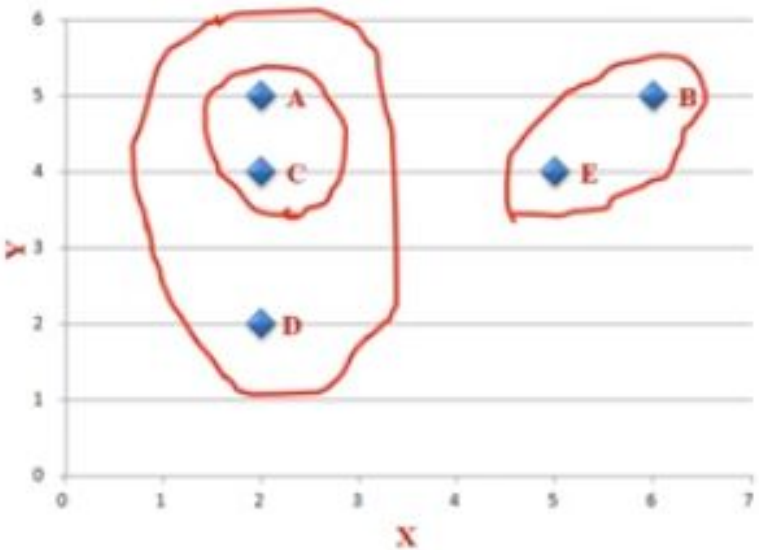
$= \min(4, 3.16, 4.12, 3) = 3$

$d_{(B,E)\to D} = \min(d_{BD}, d_{ED}) = \min(5, 3.16) = 3.16$

	A,C	B,E	D
A,C	0	3	2
B,E	3	0	3.16
D	2	3.16	0

Distance Matrix

	A,C	B,E	D
A,C	0	3	2
B,E	3	0	3.16
D	2	3.16	0



	(A,C),D	B,E
(A,C),D	0	3
B,E	3	0

$d_{(A,C),D \rightarrow (B,E)} = \min(d_{AB}, d_{AE}, d_{CB}, d_{CE}, d_{DB}, d_{DE})$

$= \min(\mathbf{4, 3.16, 4.12, 3, 5, 3.6}) = 3$

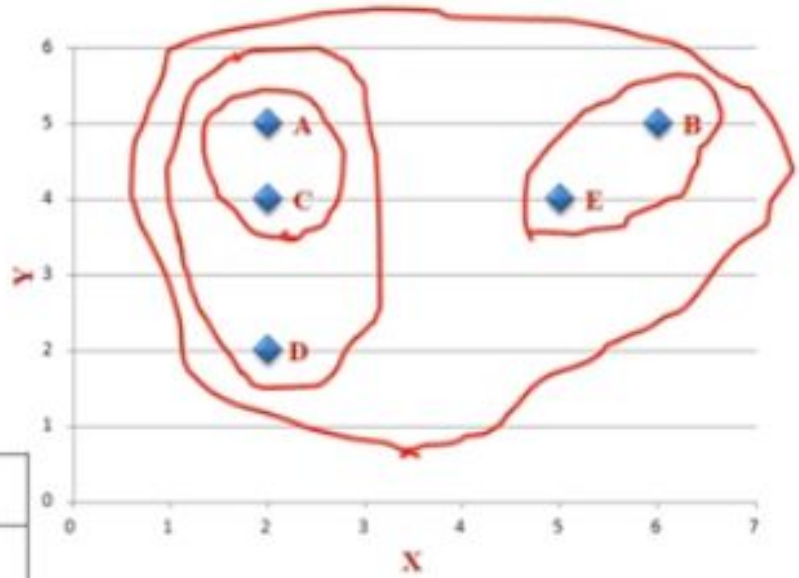
Distance Matrix

	(A,C),D	B,E
(A,C),D	0	3
B,E	3	0

((A,C),D),(B,E))

(A,B,C,D,E)

	X	Y
A	2	5
B	6	5
C	2	4
D	2	2
E	5	4

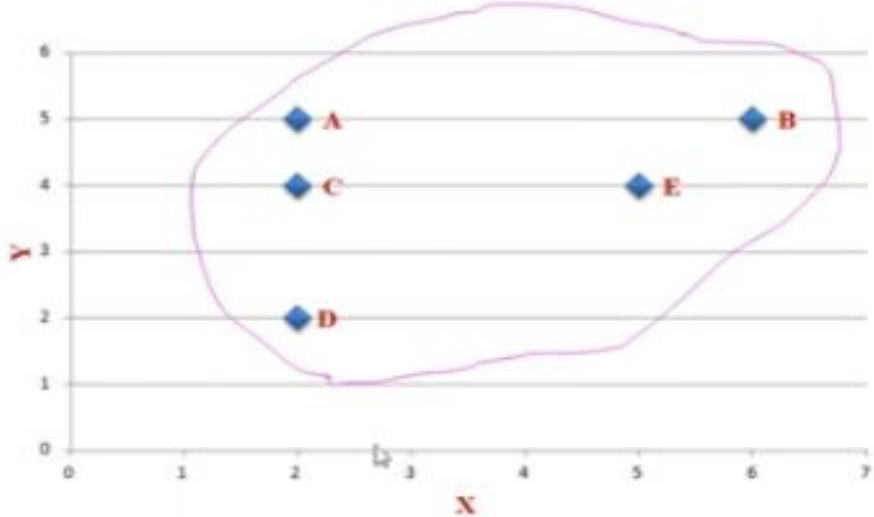


Divisive Hierarchical clustering Technique

- Divisive Hierarchical approach is commonly known as the top-down approach.
- Start with a single cluster than break it up into smaller clusters.
- Clusters are successively divided until all observation are in their own cluster.
- **Group Average:** Take all the pairs of points and compute their similarities and calculate the average of the similarities.

Example - DIANA (Divisive Analysis)

	X	Y
A	2	5
B	6	5
C	2	4
D	2	2
E	5	4



Distance Matrix (using Euclidean distance)

	A	B	C	D	E
A	0	4	1	3	3.16
B	4	0	4.12	5	1.41
C	1	4.12	0	2	3
D	3	5	2	0	3.6
E	3.16	1.41	3	3.6	0

Average dissimilarity to other objects

(4+1+ 3+3.2)/4=2.79

(4+4.12+5+1.41)/4=3.63

(1+4.12+ 2+3)/4=2.53

(3+5+ 2+3.6)/4=3.4

(3.16+1.41+ 3+3.6)/4=2.792

A	2.79
B	3.6325
C	2.53
D	3.4
E	2.7925

At this stage we have the groups {B} and {A, C, D, E}.

Remaining Point and compare it to the average dissimilarity

	A	C	D	E
A	0	1	3	3.16
C	1	0	2	3
D	3	2	0	3.6
E	3.16	3	3.6	0

Average dissimilarity to other objects

	Avg	B	(Avg-B)
A	2.387	4	**-1.61333**
C	2	4.12	**-2.12**
D	2.867	5	**-2.13333**
E	3.25	1.41	**1.843333**

(1+ 3+3.2)/3=2.387

(1+2+3)/3=2

(3+2+3.6)/3=2.87

(3.16+3+3.6)/3=3.25

At this stage we have the groups {B,E} and {A, C, D}.

Remaining Point and compare it to the average dissimilarity

	A	C	D
A	0	1	3
C	1	0	2
D	3	2	0

(1+ 3)/2=2

(1+2)/2=1.5

(3+2)/2=2.5

Average dissimilarity to other objects

	Avg	Avg(B+E)	Difference
A	2	3.6	-1.6
C	1.5	3.55	-2.05
D	2.5	4.3	-1.8

Avg(B+E)
(4+3.2)/2=3.6
(4.1+3)/2=3.55
(5+3.6)/2=4.3

Remaining Point and compare it to the average dissimilarity

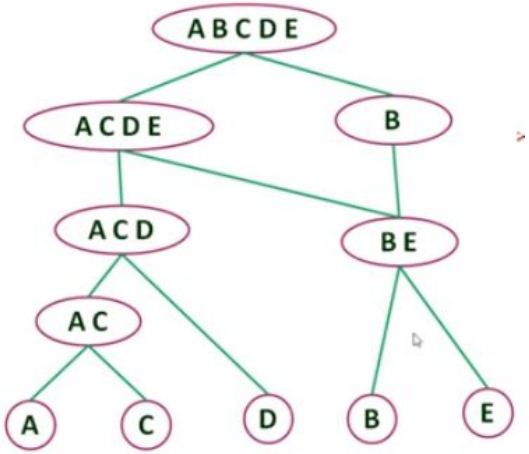
	A	C	D
A	0	1	3
C	1	0	2
D	3	2	0

Average dissimilarity to other objects

	Avg	Avg(B+E)	Difference
A	2	3.6	-1.6
C	1.5	3.55	-2.05
D	2.5	4.3	-1.8

All negative

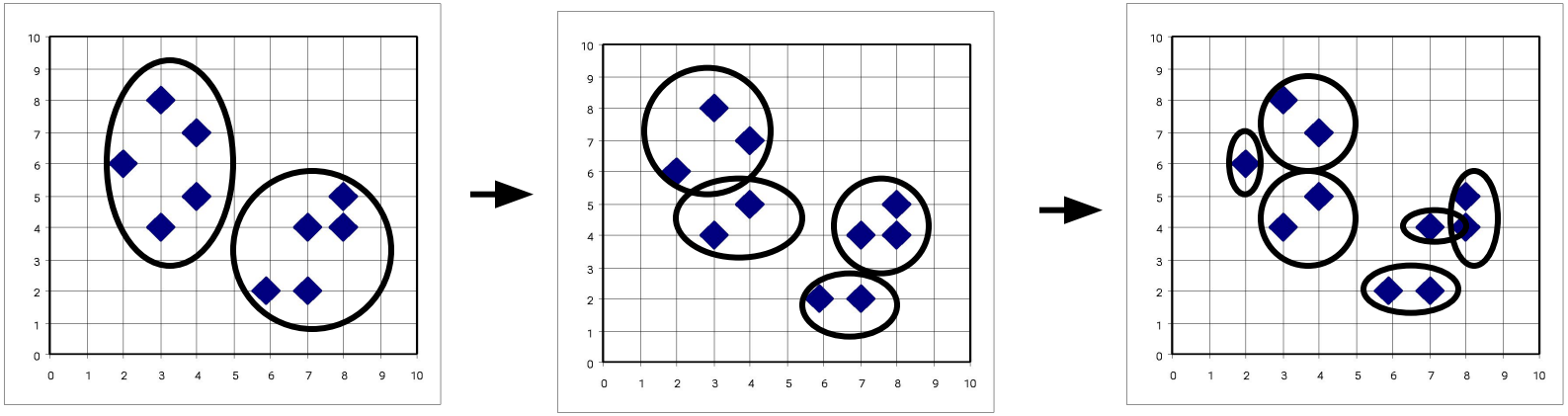
	Avg
A	2
C	1.5
D	2.5



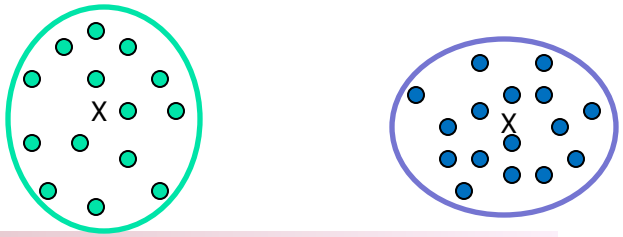
At this stage we have the groups {B,E},{D} and {A, C}.
(B) (E) (D) (A) (C)

DIANA (Divisive Analysis)

- Introduced in Kaufmann and Rousseeuw (1990)
- Implemented in statistical analysis packages, e.g., Splus
- Inverse order of AGNES
- Eventually each node forms a cluster on its own



Distance between Clusters



- **Single link:** smallest distance between an element in one cluster and an element in the other, i.e., $dist(K_i, K_j) = min(t_{ip}, t_{jq})$
- **Complete link:** largest distance between an element in one cluster and an element in the other, i.e., $dist(K_i, K_j) = max(t_{ip}, t_{jq})$
- **Average:** avg distance between an element in one cluster and an element in the other, i.e., $dist(K_i, K_j) = avg(t_{ip}, t_{jq})$
- **Centroid:** distance between the centroids of two clusters, i.e., $dist(K_i, K_j) = dist(C_i, C_j)$
- **Medoid:** distance between the medoids of two clusters, i.e., $dist(K_i, K_j) = dist(M_i, M_j)$
  - Medoid: a chosen, centrally located object in the cluster

Centroid, Radius and Diameter of a Cluster (for numerical data sets)

- Centroid: the "middle" of a cluster

$$C_m = \frac{\sum_{i=1}^{N}(t_{ip})}{N}$$

- Radius: square root of average distance from any point of the cluster to its centroid

$$R_m = \sqrt{\frac{\sum_{i=1}^{N}(t_{ip} - c_m)^2}{N}}$$

- Diameter: square root of average mean squared distance between all pairs of points in the cluster

$$D_m = \sqrt{\frac{\sum_{i=1}^{N}\sum_{i=1}^{N}(t_{ip} - t_{iq})^2}{N(N-1)}}$$

Extensions to Hierarchical Clustering

- Major weakness of agglomerative clustering methods
  - Can never undo what was done previously
  - Do not scale well: time complexity of at least $O(n^2)$, where n is the number of total objects
- Integration of hierarchical & distance-based clustering
  - BIRCH (1996): uses CF-tree and incrementally adjusts the quality of sub-clusters
  - CHAMELEON (1999): hierarchical clustering using dynamic modeling

BIRCH (Balanced Iterative Reducing and Clustering Using Hierarchies)

- Zhang, Ramakrishnan & Livny, SIGMOD'96
- Incrementally construct a CF (Clustering Feature) tree, a hierarchical data structure for multiphase clustering
  - Phase 1: scan DB to build an initial in-memory CF tree (a multi-level compression of the data that tries to preserve the inherent clustering structure of the data)
  - Phase 2: use an arbitrary clustering algorithm to cluster the leaf nodes of the CF-tree
- *Scales linearly*: finds a good clustering with a single scan and improves the quality with a few additional scans
- *Weakness:* handles only numeric data, and sensitive to the order of the data record

Clustering Feature Vector in BIRCH

Clustering Feature (CF): ***CF = (N, LS, SS)***

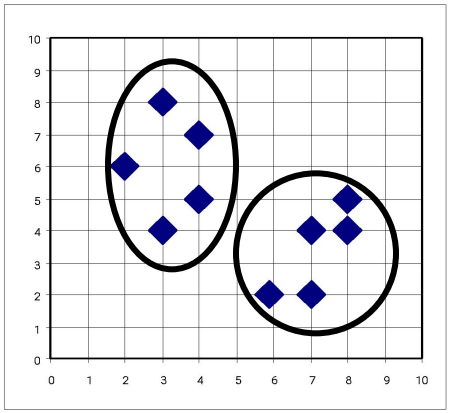
N: **Number of data points**

LS: linear sum of N points: $\sum_{i=1}^{N} X_i$

SS: square sum of N points

$$\sum_{i=1}^{N} X_i^2$$

CF = (5, (16,30),(54,190))

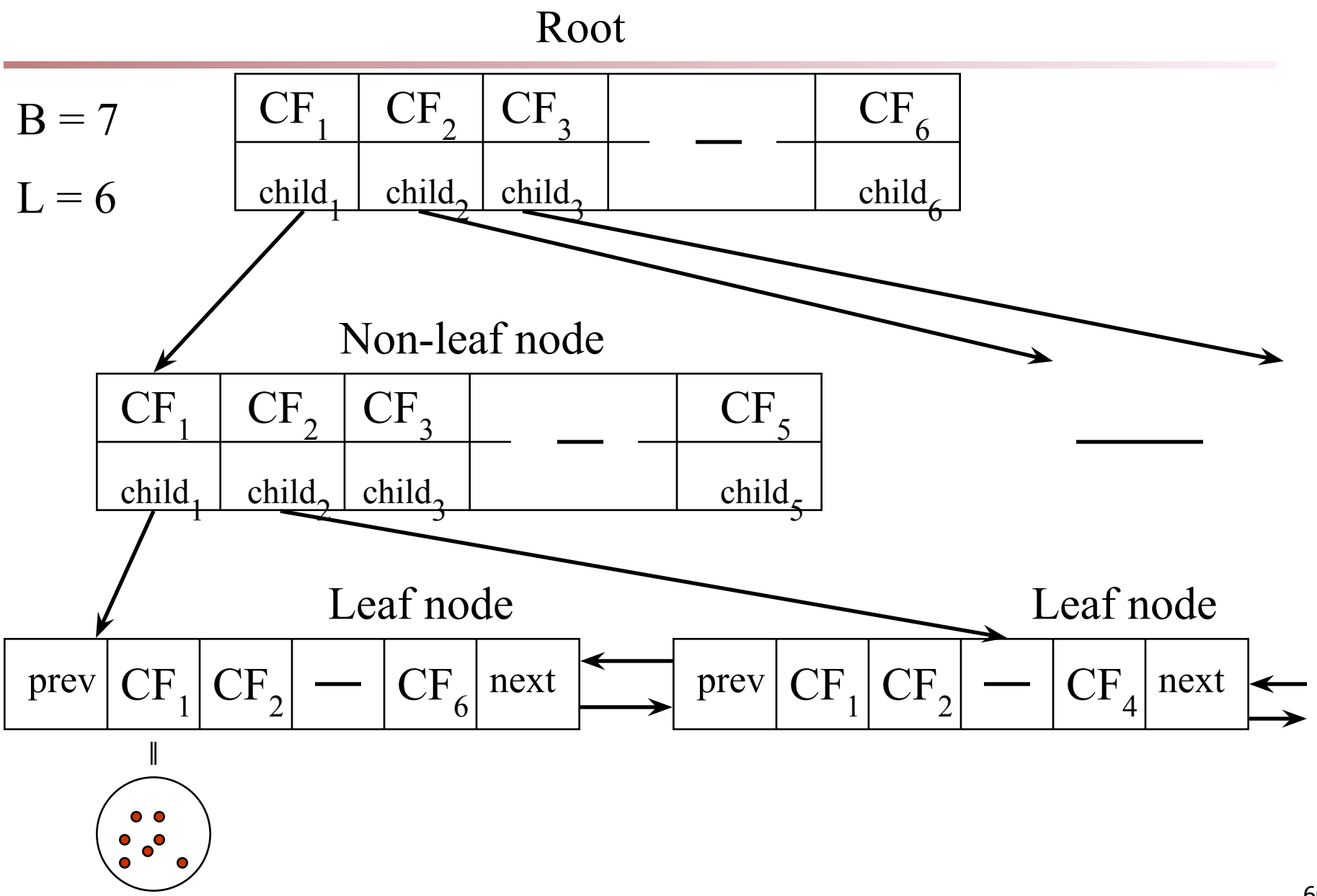


(3,4)
(2,6)
(4,5)
(4,7)
(3,8)

CF-Tree in BIRCH

- Clustering feature:
  - Summary of the statistics for a given subcluster: the 0-th, 1st, and 2nd moments of the subcluster from the statistical point of view
  - Registers crucial measurements for computing cluster and utilizes storage efficiently
- A CF tree is a height-balanced tree that stores the clustering features for a hierarchical clustering
  - A nonleaf node in a tree has descendants or “children”
  - The nonleaf nodes store sums of the CFs of their children
- A CF tree has two parameters
  - Branching factor: max # of children
  - Threshold: max diameter of sub-clusters stored at the leaf nodes

The CF Tree Structure



The Birch Algorithm

- Cluster Diameter

$$\sqrt{\frac{1}{n(n-1)}\sum(x_i - x_j)^2}$$

- For each point in the input
  - Find closest leaf entry
  - Add point to leaf entry and update CF
  - If entry diameter > max_diameter, then split leaf, and possibly parents
- Algorithm is O(n)
- Concerns
  - Sensitive to insertion order of data points
  - Since we fix the size of leaf nodes, so clusters may not be so natural
  - Clusters tend to be spherical given the radius and diameter measures

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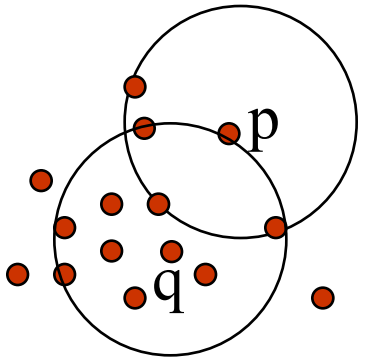
Density-Based Clustering Methods

- Clustering based on density (local cluster criterion), such as density-connected points
- Major features:
  - Discover clusters of arbitrary shape
  - Handle noise
  - One scan
  - Need density parameters as termination condition
- Several interesting studies:
  - DBSCAN: Ester, et al. (KDD'96)
  - OPTICS: Ankerst, et al (SIGMOD'99).
  - DENCLUE: Hinneburg & D. Keim (KDD'98)
  - CLIQUE: Agrawal, et al. (SIGMOD'98) (more grid-based)

Density-Based Clustering: Basic Concepts

- Two parameters*:*
  - *Eps*: Maximum radius of the neighbourhood
  - *MinPts*: Minimum number of points in an Eps-neighbourhood of that point
- $N_{Eps}(p)$: {q belongs to D | dist(p,q) ≤ Eps}
- Directly density-reachable: A point *p* is directly density-reachable from a point *q* w.r.t. *Eps*, *MinPts* if
  - *p* belongs to $N_{Eps}(q)$
  - core point condition:

$$|N_{Eps}(q)| \geq MinPts$$

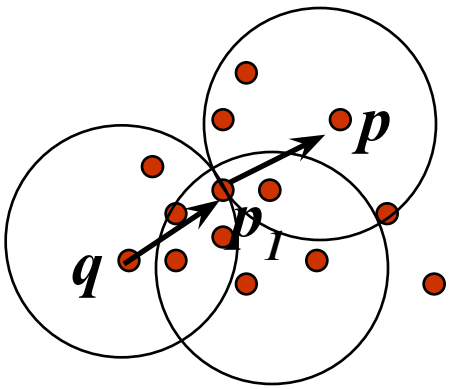


MinPts = 5

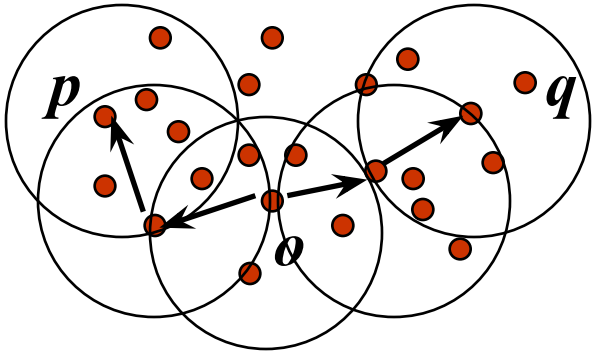
Eps = 1 cm

Density-Reachable and Density-Connected

- Density-reachable:
  - A point p is density-reachable from a point q w.r.t. *Eps*, *MinPts* if there is a chain of points $p_1, \ldots, p_n$, $p_1 = q$, $p_n = p$ such that p_{i+1} is directly density-reachable from p_i



- Density-connected
  - A point p is density-connected to a point q w.r.t. *Eps*, *MinPts* if there is a point o such that both, p and q are density-reachable from o w.r.t. *Eps* and *MinPts*



DBSCAN

- Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

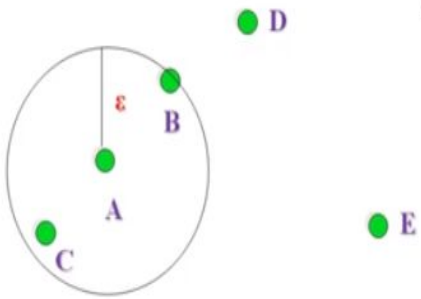
- It is able to find arbitrary shaped clusters and clusters with noise (i.e. outliers).

DBSCAN categories the data points into three categories:

1. Core point
2. Border point
3. Noise point

ε= 1cm

MinPts = 3

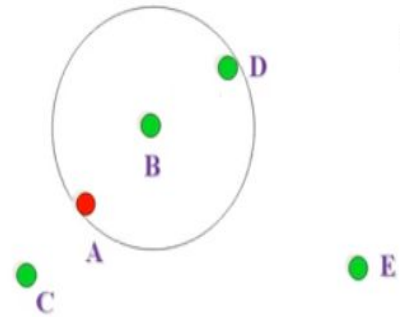


DBSCAN categories the data points into three categories:

1. Core point ●
2. Border point
3. Noise point

ε= 1cm

MinPts = 3



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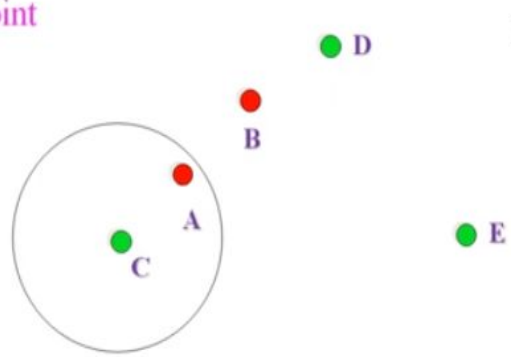
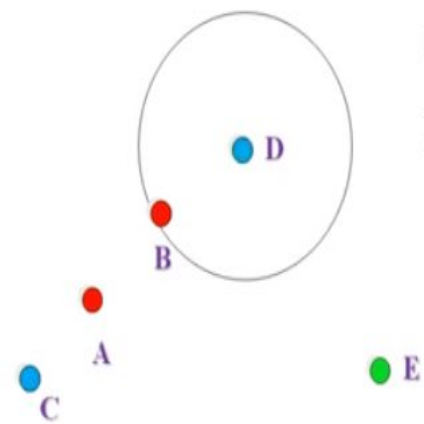


DBSCAN categories the data points into three categories:

1. Core point ●
2. Border point ●
3. Noise point

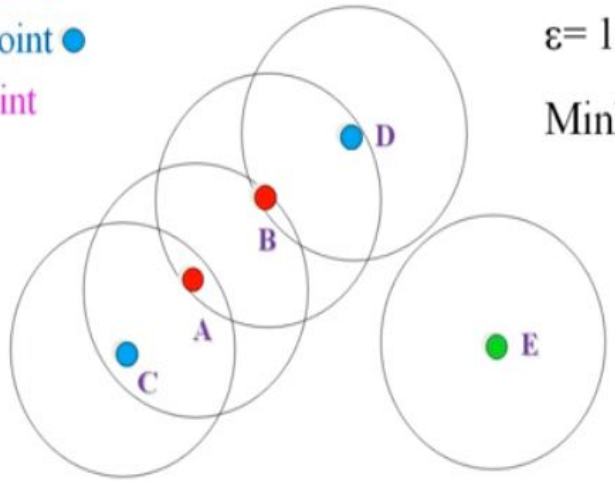
ε= 1cm

MinPts = 3



DBSCAN categories the data points into three categories:

1. Core point ●
2. Border point ●
3. Noise point



ε= 1cm

MinPts = 3

DBSCAN Algorithm

Input: The data set D

Parameter: ε, MinPts

For each object p in D

if p is a core object and not processed then

C = retrieve all objects density-reachable from p

mark all objects in C as processed

report C as a cluster

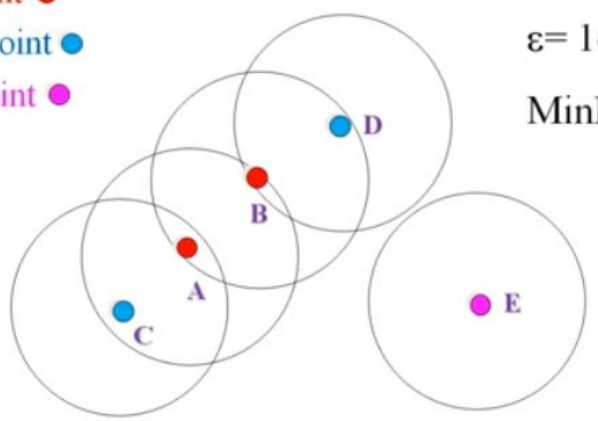
else mark p as outlier

end if

End For

DBSCAN categories the data points into three categories:

1. Core point ●
2. Border point ●
3. Noise point ●



ε= 1cm

MinPts = 3

- Apply the DBSCAN algorithm to the given data points and
- Create the clusters with
- minPts = 4 and
- epsilon (ε) = 1.9.

Data Points:

P1: (3, 7)	P2: (4, 6)
P3: (5, 5)	P4: (6, 4)
P5: (7, 3)	P6: (6, 2)
P7: (7, 2)	P8: (8, 4)
P9: (3, 3)	P10: (2, 6)
P11: (3, 5)	P12: (2, 4)

- Use Eucladian distance and calculate the distance between each points.

$$\boldsymbol{Distance}(\boldsymbol{A}(\boldsymbol{x_1}, \boldsymbol{y_1}), \boldsymbol{B}(\boldsymbol{x_2}, \boldsymbol{y_2})) = \sqrt{(\boldsymbol{x_2} - \boldsymbol{x_1})^2 + (\boldsymbol{y_2} - \boldsymbol{y_1})^2}$$

P1: (3, 7)
P2: (4, 6)
P3: (5, 5)
P4: (6, 4)
P5: (7, 3)
P6: (6, 2)
P7: (7, 2)
P8: (8, 4)
P9: (3, 3)
P10: (2, 6)
P11: (3, 5)
P12: (2, 4)

minPts = 4 and epsilon (ε) = 1.9

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2												
P3												
P4												
P5												
P6												
P7												
P8												
P9												
P10												
P11												
P12												

P1: (3, 7)
P2: (4, 6)
P3: (5, 5)
P4: (6, 4)
P5: (7, 3)
P6: (6, 2)
P7: (7, 2)
P8: (8, 4)
P9: (3, 3)
P10: (2, 6)
P11: (3, 5)
P12: (2, 4)

minPts = 4 and epsilon (ε) = 1.9

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2	1.41											
P3	2.83 ✓											
P4	4.24											
P5	5.66											
P6	5.83											
P7	6.40											
P8	5.83											
P9	4.00											
P10	1.41											
P11	2.00											
P12	3.16											

P1: (3, 7)
P2: (4, 6)
P3: (5, 5)
P4: (6, 4)
P5: (7, 3)
P6: (6, 2)
P7: (7, 2)
P8: (8, 4)
P9: (3, 3)
P10: (2, 6)
P11: (3, 5)
P12: (2, 4)

minPts = 4 and epsilon (ε) = 1.9

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2	1.41	0										
P3	2.83	1.41	0									
P4	4.24	2.83	1.41	0								
P5	5.66	4.24	2.83	1.41	0							
P6	5.83	4.47	3.16	2.00	1.41	0						
P7	6.40	5.00	3.61	2.24	1.00	1.00	0					
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0				
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
P11	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0	
P12	3.16	2.83	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0

minPts = 4 and epsilon (ε) = 1.9

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2	1.41	0										
P3	2.83	1.41	0									
P4	4.24	2.83	1.41	0								
P5	5.66	4.24	2.83	1.41	0							
P6	5.83	4.47	3.16	2.00	1.41	0						
P7	6.40	5.00	3.61	2.24	1.00	1.00	0					
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0				
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
P11	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0	
P12	3.16	2.83	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0

P1: P2, P10

P2: P1, P3, P11

minPts = 4 and epsilon (ε) = 1.9

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2	1.41	0										
P3	2.83	1.41	0									
P4	4.24	2.83	1.41	0								
P5	5.66	4.24	2.83	1.41	0							
P6	5.83	4.47	3.16	2.00	1.41	0						
P7	6.40	5.00	3.61	2.24	1.00	1.00	0					
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0				
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
P11	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0	
P12	3.16	2.83	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0

P1: P2, P10

P2: P1, P3, P11

P3: P2, P4

P4: P3, P5

P5: P4, P6, P7, P8

P6: P5, P7

P7: P5, P6

P8: P5

P9: P12

P10: P1, P11

P11: P2, P10, P12

P12: P9, P11

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P2: P1, P3, P11

P3: P2, P4

P4: P3, P5

P5: P4, P6, P7, P8

P6: P5, P7

P7: P5, P6

P8: P5

P9: P12

P10: P1, P11

P11: P2, P10, P12

P12: P9, P11

minPts = 4 and epsilon (ε) = 1.9

Point	Status	
P1	Noise	
P2	Core	
P3	Noise	
P4	Noise	
P5	Core	
P6	Noise	
P7	Noise	
P8	Noise	
P9	Noise	
P10	Noise	
P11	Core	
P12	Noise	

P1: P2, P10

P2: P1, P3, P11

P3: P2, P4

P4: P3, P5

P5: P4, P6, P7, P8

P6: P5, P7

P7: P5, P6

P8: P5

P9: P12

P10: P1, P11

P11: P2, P10, P12

P12: P9, P11

minPts = 4 and epsilon (ε) = 1.9

Point	Status	
P1	Noise	Border
P2	Core	
P3	Noise	Border
P4	Noise	Border
P5	Core	
P6	Noise	Border
P7	Noise	Border
P8	Noise	Border
P9	Noise	
P10	Noise	Border
P11	Core	
P12	Noise	Border

P1: (3, 7)

P2: (4, 6)

P3: (5, 5)

P4: (6, 4)

P5: (7, 3)

P6: (6, 2)

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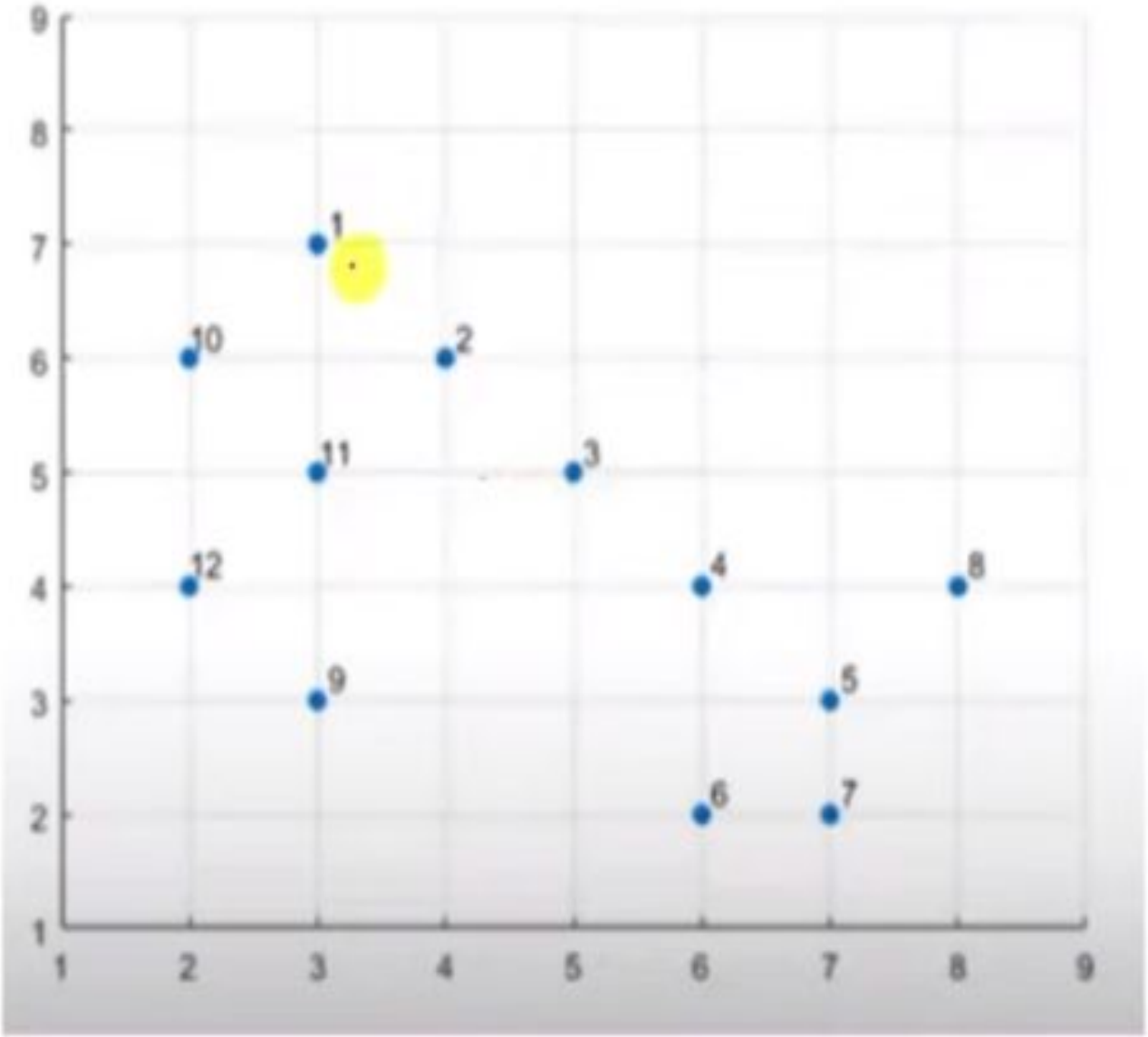
minPts = 4 and epsilon (ε) = 1.9



P1: P2, P10

P2: P1, P3, P11

P3: P2, P4

P4: P3, P5

P5: P4, P6, P7, P8

P6: P5, P7

P7: P5, P6

P8: P5

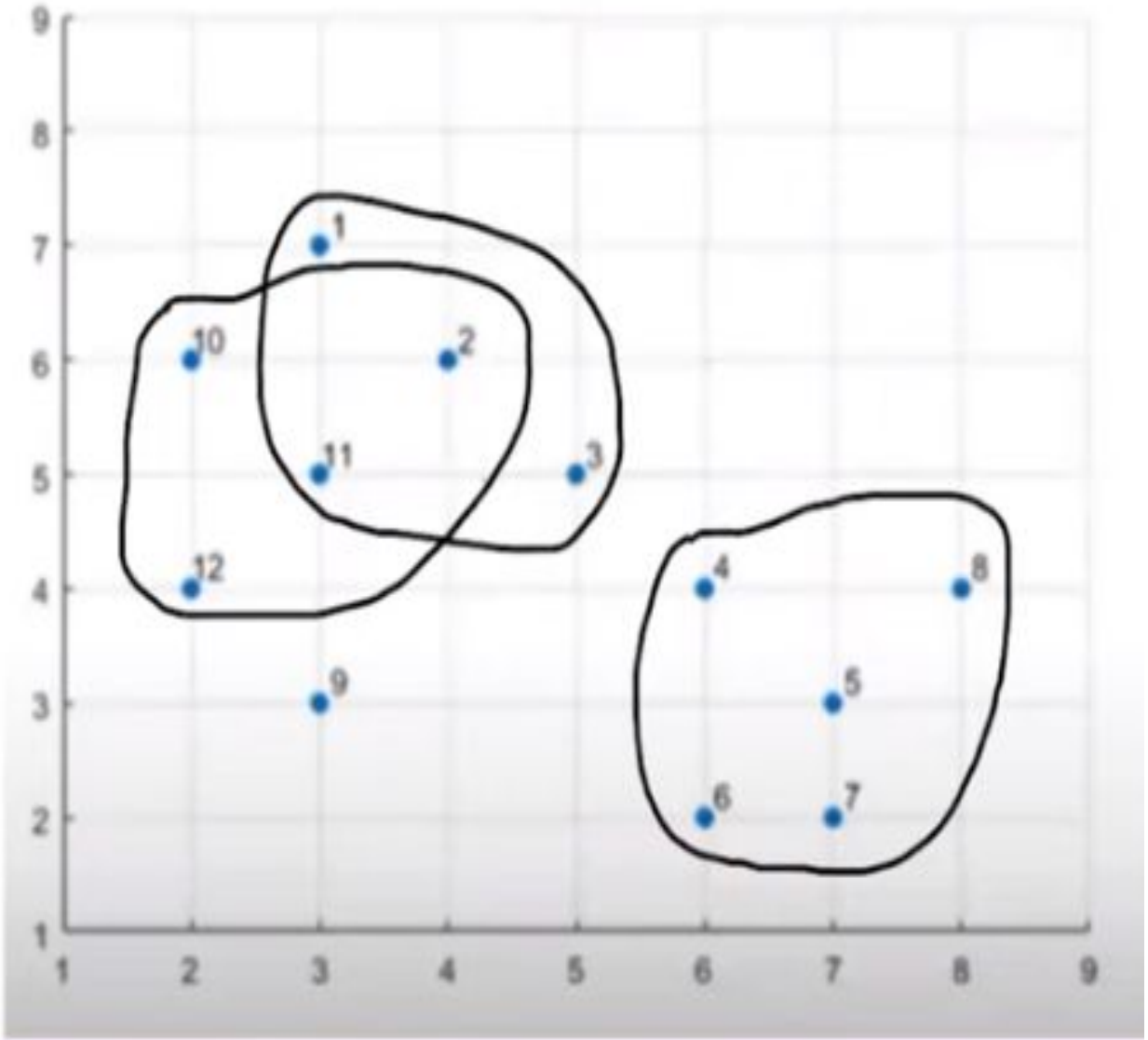
P9: P12

P10: P1, P11

P11: P2, P10, P12

P12: P9, P11

minPts = 4 and epsilon (ε) = 1.9



Advantages

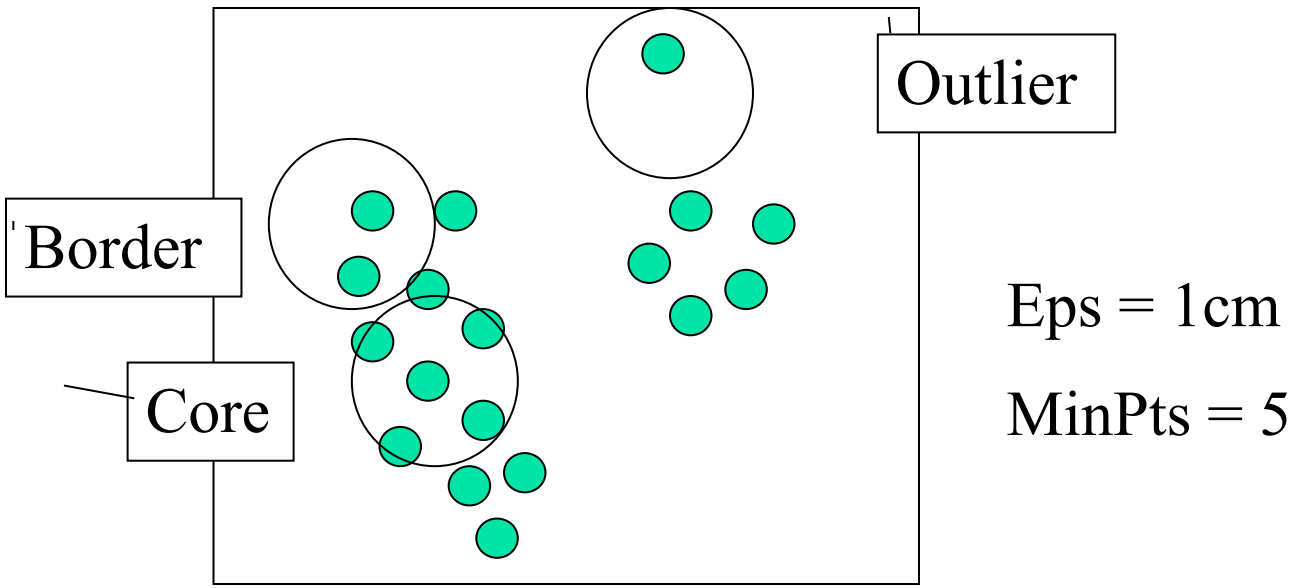
1) Does not require a-priori specification of number of clusters.
2) Able to identify noise data while clustering.
3) DBSCAN algorithm is able to find arbitrarily size and arbitrarily shaped clusters.

Disadvantages

1) DBSCAN algorithm fails in case of varying density clusters.
2) Fails in case of neck type of dataset.

DBSCAN: Density-Based Spatial Clustering of Applications with Noise

- Relies on a *density-based* notion of cluster: A *cluster* is defined as a maximal set of density-connected points
- Discovers clusters of arbitrary shape in spatial databases with noise



Chapter 10. Cluster Analysis: Basic Concepts and Methods

- Cluster Analysis: Basic Concepts
- Partitioning Methods
- Hierarchical Methods
- Density-Based Methods
- Grid-Based Methods
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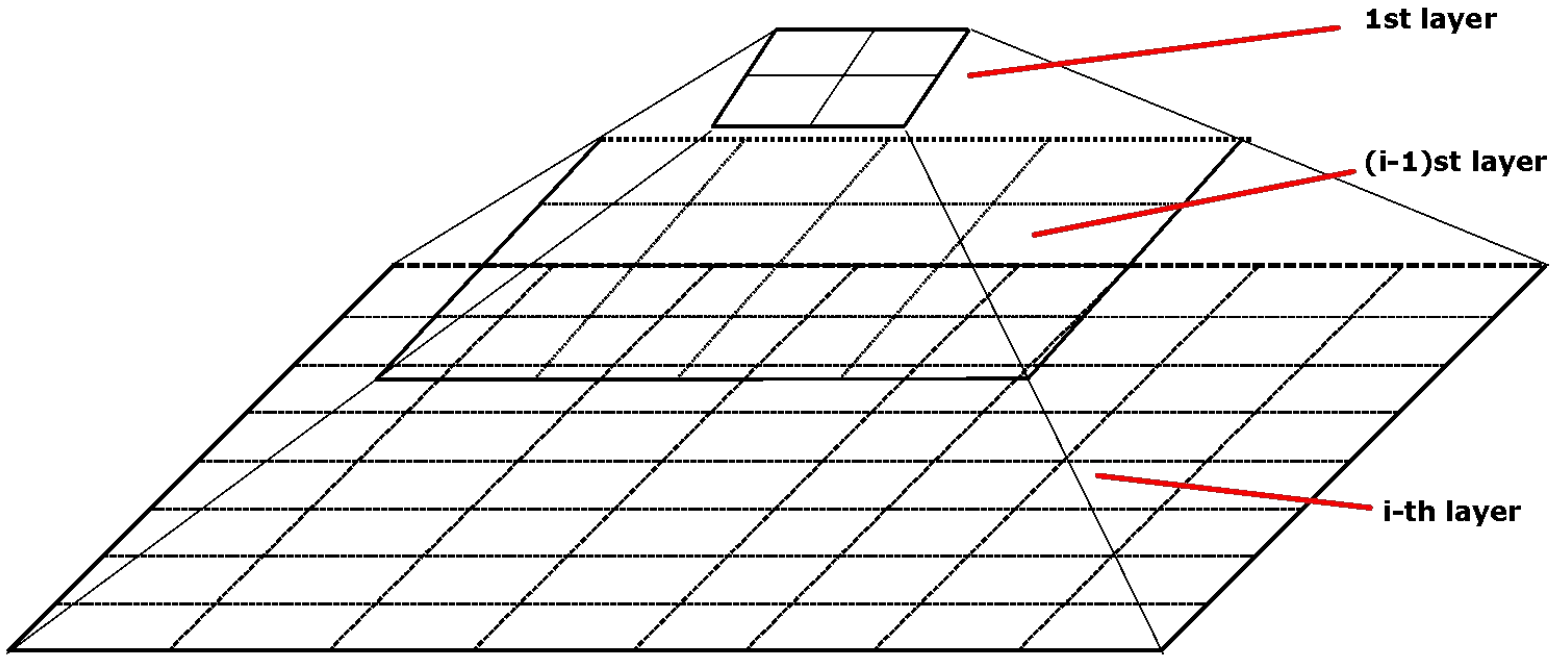
Grid-Based Clustering Method

- Using multi-resolution grid data structure
- Several interesting methods
  - STING (a STatistical INformation Grid approach)
  - CLIQUE: (Clustering In QUEst)
    - Both grid-based and subspace clustering

STING: A Statistical Information Grid Approach

- Wang, Yang and Muntz (VLDB'97)
- The spatial area is divided into rectangular cells
- There are several levels of cells corresponding to different levels of resolution



The STING Clustering Method

- Each cell at a high level is partitioned into a number of smaller cells in the next lower level
- Statistical info of each cell is calculated and stored before hand and is used to answer queries
- Parameters of higher level cells can be easily calculated from parameters of lower level cell
  - *count*, *mean*, *sd*, *min*, *max*
  - type of distribution—*normal*, *uniform*, etc.
- Use a top-down approach to answer spatial data queries
- Start from a pre-selected layer—typically with a small number of cells
- For each cell in the current level compute the confidence interval

STING Algorithm and Its Analysis

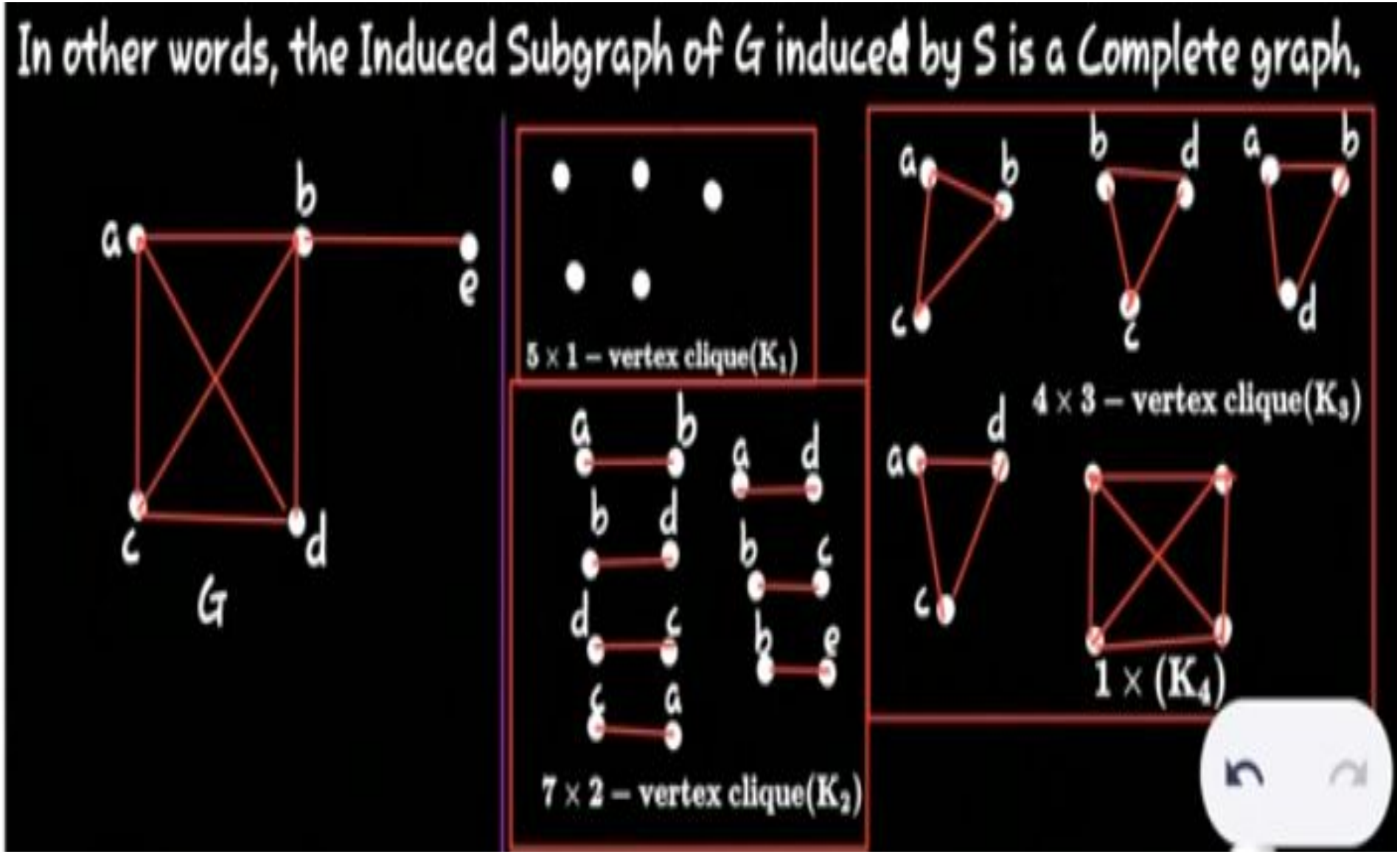
- Remove the irrelevant cells from further consideration
- When finish examining the current layer, proceed to the next lower level
- Repeat this process until the bottom layer is reached
- Advantages:
  - Query-independent, easy to parallelize, incremental update
  - *O(K),* where *K* is the number of grid cells at the lowest level
- Disadvantages:
  - All the cluster boundaries are either horizontal or vertical, and no diagonal boundary is detected

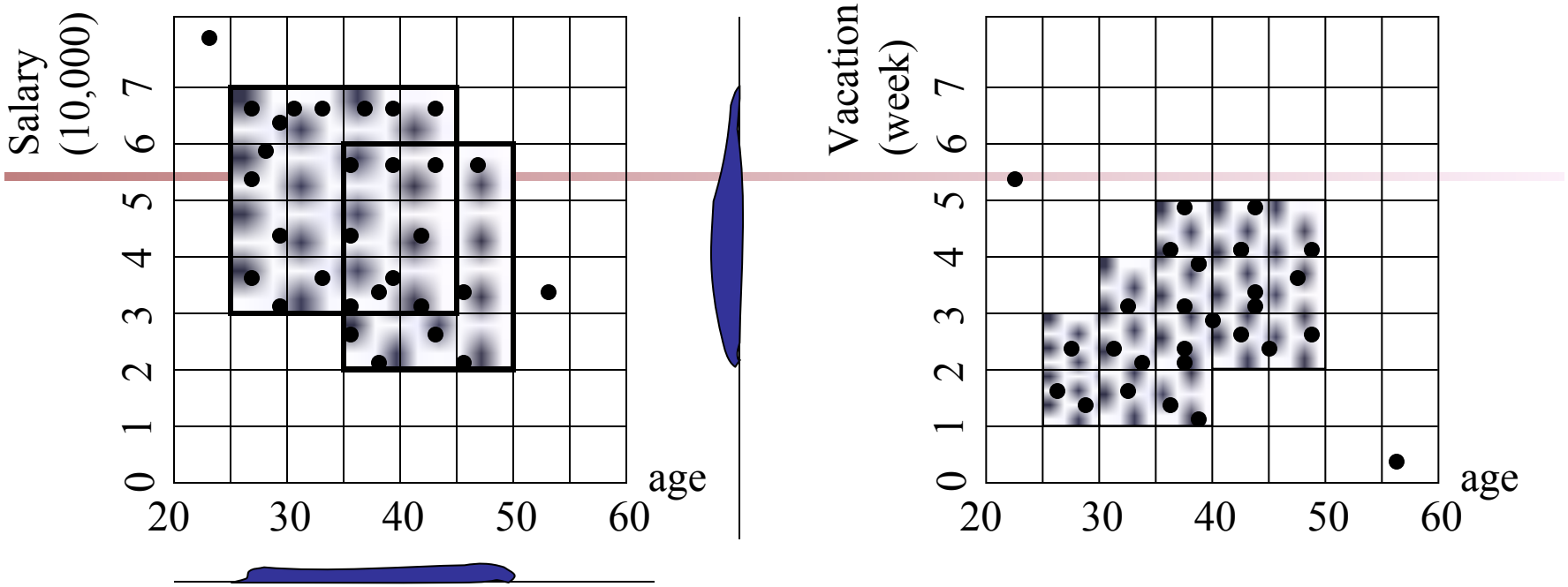
CLIQUE (Clustering In QUEst)

- Agrawal, Gehrke, Gunopulos, Raghavan (SIGMOD'98)
- Automatically identifying subspaces of a high dimensional data space that allow better clustering than original space
- CLIQUE can be considered as both density-based and grid-based
  - It partitions each dimension into the same number of equal length interval
  - It partitions an m-dimensional data space into non-overlapping rectangular units
  - A unit is dense if the fraction of total data points contained in the unit exceeds the input model parameter
  - A cluster is a maximal set of connected dense units within a subspace

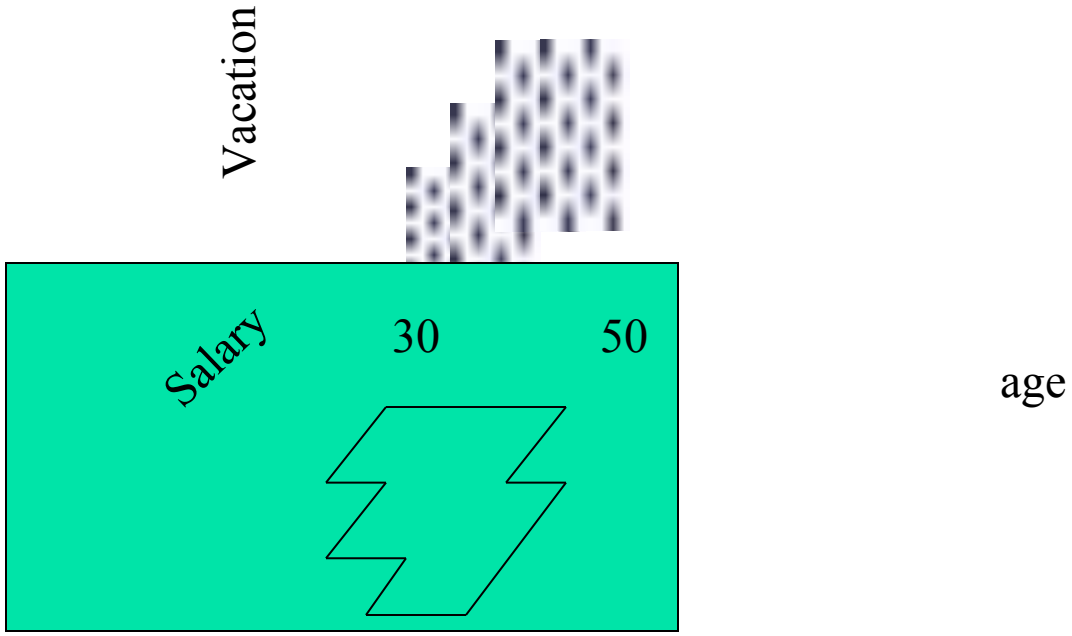
CLIQUE: The Major Steps

- Partition the data space and find the number of points that lie inside each cell of the partition.
- Identify the subspaces that contain clusters using the Apriori principle
- Identify clusters
  - Determine dense units in all subspaces of interests
  - Determine connected dense units in all subspaces of interests.
- Generate minimal description for the clusters
  - Determine maximal regions that cover a cluster of connected dense units for each cluster
  - Determination of minimal cover for each cluster


In other words, the Induced Subgraph of G induced by S is a Complete graph.
a
b
c
d
e
G
5 × 1 – vertex clique(K1)
7 × 2 – vertex clique(K2)
4 × 3 – vertex clique(K3)
1 × (K4)


Salary
(10,000)
0 1 2 3 4 5 6 7
20 30 40 50 60
age
Vacation
(week)
0 1 2 3 4 5 6 7
20 30 40 50 60
age

$\tau = 3$


Vacation
Salary
30
50
age

Strength and Weakness of *CLIQUE*

- Strength
  - *automatically* finds subspaces of the highest dimensionality such that high density clusters exist in those subspaces
  - *insensitive* to the order of records in input and does not presume some canonical data distribution
  - scales *linearly* with the size of input and has good scalability as the number of dimensions in the data increases
- Weakness
  - The accuracy of the clustering result may be degraded at the expense of simplicity of the method

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Assessing Clustering Tendency

- Assess if non-random structure exists in the data by measuring the probability that the data is generated by a uniform data distribution
- Test spatial randomness by statistic test: Hopkins Static
  - Given a dataset D regarded as a sample of a random variable o, determine how far away o is from being uniformly distributed in the data space
  - Sample *n* points, $p_1, \ldots, p_n$, uniformly from D. For each p_i, find its nearest neighbor in D: $x_i = min\{dist\ (p_i, v)\}$ where *v* in D
  - Sample *n* points, $q_1, \ldots, q_n$, uniformly from D. For each q_i, find its nearest neighbor in D – $\{q_i\}$: $y_i = min\{dist\ (q_i, v)\}$ where *v* in D and $v \neq q_i$
  - Calculate the Hopkins Statistic: $H = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i}$
  - If D is uniformly distributed, $\sum x_i$ and $\sum y_i$ will be close to each other and H is close to 0.5. If D is highly skewed, H is close to 0

Determine the Number of Clusters

- Empirical method
  - # of clusters ≈$\sqrt{n/2}$ for a dataset of n points
- Elbow method
  - Use the turning point in the curve of sum of within cluster variance w.r.t the # of clusters
- Cross validation method
  - Divide a given data set into *m* parts
  - Use *m* – 1 parts to obtain a clustering model
  - Use the remaining part to test the quality of the clustering
    - E.g., For each point in the test set, find the closest centroid, and use the sum of squared distance between all points in the test set and the closest centroids to measure how well the model fits the test set
  - For any k > 0, repeat it *m* times, compare the overall quality measure w.r.t. different *k's*, and find # of clusters that fits the data the best

Measuring Clustering Quality

- Two methods: extrinsic vs. intrinsic
- Extrinsic: supervised, i.e., the ground truth is available
  - Compare a clustering against the ground truth using certain clustering quality measure
  - Ex. BCubed precision and recall metrics
- Intrinsic: unsupervised, i.e., the ground truth is unavailable
  - Evaluate the goodness of a clustering by considering how well the clusters are separated, and how compact the clusters are
  - Ex. Silhouette coefficient

Measuring Clustering Quality: Extrinsic Methods

- Clustering quality measure: $Q(C, C_g)$, for a clustering C given the ground truth C_g.
- Q is good if it satisfies the following **4** essential criteria
  - Cluster homogeneity: the purer, the better
  - Cluster completeness: should assign objects belong to the same category in the ground truth to the same cluster
  - Rag bag: putting a heterogeneous object into a pure cluster should be penalized more than putting it into a *rag bag* (i.e., “miscellaneous” or “other” category)
  - Small cluster preservation: splitting a small category into pieces is more harmful than splitting a large category into pieces

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Summary

- Cluster analysis groups objects based on their similarity and has wide applications
- Measure of similarity can be computed for various types of data
- Clustering algorithms can be categorized into partitioning methods, hierarchical methods, density-based methods, grid-based methods, and model-based methods
- K-means and K-medoids algorithms are popular partitioning-based clustering algorithms
- Birch and Chameleon are interesting hierarchical clustering algorithms, and there are also probabilistic hierarchical clustering algorithms
- DBSCAN, OPTICS, and DENCLU are interesting density-based algorithms
- STING and CLIQUE are grid-based methods, where CLIQUE is also a subspace clustering algorithm
- Quality of clustering results can be evaluated in various ways